Introduction to Harmonic Analysis

Lecture notes, expanded

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Abstract

These notes follow the handwritten notebook page by page; the grey rules "notebook page N" mark where page N of the notebook begins. Everything said in the lectures is here, but in addition every notion that was used without definition is defined (in grey *Background* boxes), every "check this" and every exercise is worked out, and remarks are explained.

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1 Fourier analysis in Hilbert spaces

1.1 Inner products and norms

Background 1. (Vector spaces over $\mathbb{C}$)

A *linear space* (vector space) over $\mathbb{C}$ is a set H with an addition $H \times H \to H$ and a scalar multiplication $\mathbb{C} \times H \to H$ such that $(H, +)$ is an abelian group (with neutral element 0) and for all $\alpha, \beta \in \mathbb{C}$, $g, h \in H$:

$$\alpha(g+h) = \alpha g + \alpha h, \quad (\alpha+\beta)g = \alpha g + \beta g, \quad (\alpha\beta)g = \alpha(\beta g), \quad 1 \cdot g = g.$$

In particular $g, h \in H$, $\alpha, \beta \in \mathbb{C} \Rightarrow \alpha g + \beta h \in H$. For $z = a + ib \in \mathbb{C}$ we write $\overline{z} = a - ib$, $\operatorname{Re} z = a = \frac{z+\overline{z}}{2}$, $\operatorname{Im} z = b = \frac{z-\overline{z}}{2i}$ and $|z|^2 = z\overline{z}$.

Definition 1.2 (Inner product)**.** Let H be a linear space over $\mathbb{C}$. An *inner product* on H is a map $(\cdot,\cdot) : H \times H \to \mathbb{C}$ such that

(i) $(x,y) = \overline{(y,x)}$ for all $x, y \in H$ (Hermitian symmetry)

(ii) $(x,x) \geq 0$ for all $x \in H$, and $(x,x) = 0 \iff x = 0$ (positive definiteness)

(iii) $(\alpha x + \beta y, z) = \alpha\,(x,z) + \beta\,(y,z)$ for all $\alpha, \beta \in \mathbb{C}$, $x, y, z \in H$ (linearity in the first slot)

Note that (i) implies $(x,x) = \overline{(x,x)} \in \mathbb{R}$, so the inequality in (ii) makes sense.

Remark 1.3 (Conjugate linearity in the second slot). $(x, \alpha y + \beta z) = \overline{\alpha}\,(x,y) + \overline{\beta}\,(x,z)$. Indeed, by (i), (iii), (i): $(x, \alpha y + \beta z) = \overline{(\alpha y + \beta z, x)} = \overline{\alpha\,(y,x) + \beta\,(z,x)} = \overline{\alpha}\,\overline{(y,x)} + \overline{\beta}\,\overline{(z,x)} = \overline{\alpha}\,(x,y) + \overline{\beta}\,(x,z)$. (Physicists use the opposite convention: linear in the second slot.)

Definition 1.4. For $x \in H$ put $\|x\| := \sqrt{(x,x)}$.

Background 5. (Norm)

A *norm* on a linear space H is a map $\|\cdot\| : H \to [0,\infty)$ with

(N1) $\|x\| = 0 \iff x = 0$;

(N2) $\|\lambda x\| = |\lambda|\,\|x\|$ for $\lambda \in \mathbb{C}$;

(N3) $\|x + y\| \leq \|x\| + \|y\|$ (triangle inequality).

A norm induces a metric $d(x,y) = \|x - y\|$, hence notions of convergence ($x_n \to x$ iff $\|x_n - x\| \to 0$), open and closed sets, closure, etc.

Proposition 1.6. *$\|\cdot\|$ is a norm on H.*

The key step is the following inequality, which we will use constantly.

Theorem 1.7 (Cauchy–Schwarz inequality)**.** *For all $x, y \in H$,*

$$|(x,y)| \leq \|x\|\,\|y\|\,. \tag{1}$$

Equality holds if and only if x and y are linearly dependent.

Proof. If $(x,y) = 0$ there is nothing to prove; so let $(x,y) \neq 0$, in particular $x \neq 0 \neq y$. For every $\alpha \in \mathbb{C}$,

$$\begin{aligned} 0 \leq \|x + \alpha y\|^2 &= (x + \alpha y, x + \alpha y) = (x,x) + (x, \alpha y) + (\alpha y, x) + |\alpha|^2\,(y,y) \\ &= \|x\|^2 + |\alpha|^2\,\|y\|^2 + 2\,\mathrm{Re}\,(x, \alpha y)\,, \end{aligned}$$

because $(\alpha y, x) = \overline{(x, \alpha y)}$ and $w + \overline{w} = 2\,\mathrm{Re}\,w$. Choose α with $|\alpha| = 1$ such that $(x, \alpha y) = \overline{\alpha}\,(x,y) = -\,|(x,y)|$, i.e.

$$\overline{\alpha} = -\frac{|(x,y)|}{(x,y)}, \qquad \alpha = -\frac{|(x,y)|}{\overline{(x,y)}} = -\frac{|(x,y)|}{(y,x)}.$$

Then $0 \leq \|x\|^2 + \|y\|^2 - 2\,|(x,y)|$, that is

$$2\,|(x,y)| \leq \|x\|^2 + \|y\|^2 \qquad \forall x, y \in H. \tag{2}$$

This is weaker than (1) but it is not homogeneous, and we can exploit that: apply (2) to x and ty, $t > 0$:

$$2t\,|(x,y)| \leq \|x\|^2 + t^2\,\|y\|^2\,.$$

Choosing $t = \|x\| / \|y\|$ (so that $t^2 \|y\|^2 = \|x\|^2$) gives $2t\,|(x,y)| \le 2\|x\|^2$, i.e. $|(x,y)| \le \frac{1}{t}\|x\|^2 = \frac{\|y\|}{\|x\|}\|x\|^2 = \|x\|\,\|y\|$.

Equality. If $y = \lambda x$ (or $x = 0$) equality is clear. Conversely, if $|(x,y)| = \|x\|\,\|y\|$ with $x, y \neq 0$, then for $t = \|x\| / \|y\|$ and the α above, $\|x + \alpha t y\|^2 = \|x\|^2 + t^2\|y\|^2 - 2t\,|(x,y)| = 0$, so $x = -\alpha t y$. □

Proof of Proposition 1.6. (N1) is (ii) of Definition 1.2. (N2): $\|\lambda x\|^2 = (\lambda x, \lambda x) = \lambda\overline{\lambda}\,(x,x) = |\lambda|^2\|x\|^2$. (N3): using $\operatorname{Re} w \le |w|$ and (1),

$$\|x+y\|^2 = \|x\|^2 + 2\operatorname{Re}(x,y) + \|y\|^2 \le \|x\|^2 + 2\|x\|\,\|y\| + \|y\|^2 = \big(\|x\| + \|y\|\big)^2. \qquad \square$$

Remark 1.8. Two identities that follow by simply expanding $\|x \pm y\|^2$ are worth remembering:

$$\|x+y\|^2 + \|x-y\|^2 = 2\|x\|^2 + 2\|y\|^2 \qquad \text{(parallelogram law)},$$
$$\|x+y\|^2 = \|x\|^2 + \|y\|^2 \quad \text{if } (x,y) = 0 \qquad \text{(Pythagoras)}.$$

The parallelogram law characterises norms coming from an inner product (Jordan–von Neumann theorem); e.g. $\max_i |x_i|$ on $\mathbb{C}^2$ does not satisfy it.

1.2 Hilbert spaces

Background 9. (Cauchy sequences and completeness)

A sequence (x_n) in a normed space is a *Cauchy sequence* if

$$\forall \varepsilon > 0\ \exists N_\varepsilon\ \forall m, n \ge N_\varepsilon :\ \|x_n - x_m\| < \varepsilon.$$

Every convergent sequence is Cauchy ($\|x_n - x_m\| \le \|x_n - x\| + \|x - x_m\|$). The space is *complete* if conversely every Cauchy sequence converges, i.e. there is x in the space with $\|x_n - x\| \to 0$. A complete normed space is called a *Banach space*. Example: $\mathbb{R}$ and $\mathbb{C}$ are complete, $\mathbb{Q}$ is not ($1,\ 1.4,\ 1.41,\ 1.414, \cdots \to \sqrt{2} \notin \mathbb{Q}$).

Definition 1.10 (Hilbert space)**.** A *Hilbert space* is a linear space H with an inner product $(\cdot,\cdot)$ such that H is complete as a normed space $(H, \|\cdot\|)$, $\|x\| = \sqrt{(x,x)}$: for every Cauchy sequence $(x_n) \subset H$ there exists $x \in H$ with $\|x_n - x\| \xrightarrow{n\to\infty} 0$.

Example 1.11 (An inner product space which is not complete)**.** On $C([0,1])$ consider $(f,g) = \int_0^1 f\overline{g}\,dx$. Let f_n be 0 on $[0, \frac{1}{2}]$, 1 on $[\frac{1}{2} + \frac{1}{n}, 1]$ and linear in between. Then $\|f_n - f_m\|^2 \le \frac{1}{\min(m,n)}$, so (f_n) is Cauchy, but its L^2-limit is the indicator of $(\frac{1}{2}, 1]$, which is not (a.e. equal to) a continuous function. So completeness is a genuine requirement; $L^2([0,1])$ below is the "completion" of this space.

Example 1.12. The following are Hilbert spaces.

(a) $\mathbb{C}^N := \left\{ a = \begin{pmatrix} a_1 \\ \vdots \\ a_N \end{pmatrix} : a_k \in \mathbb{C} \right\}$ with $(a,b)_{\mathbb{C}^N} = \sum_{k=1}^N a_k \overline{b_k}$. Completeness: a Cauchy sequence in $\mathbb{C}^N$ is Cauchy in each coordinate, and $\mathbb{C}$ is complete.

(b) $\ell^2(\mathbb{Z}) := \left\{ \{a_k\}_{k\in\mathbb{Z}} : \sum_{k\in\mathbb{Z}} |a_k|^2 < \infty \right\}$ with $(\{a_k\},\{b_k\})_{\ell^2(\mathbb{Z})} = \sum_{k\in\mathbb{Z}} a_k\overline{b_k}$.

Why this is well defined: $|a_k\overline{b_k}| \le \frac{1}{2}(|a_k|^2 + |b_k|^2)$, so the series converges absolutely; the same inequality shows $\ell^2(\mathbb{Z})$ is closed under addition. *Completeness (sketch):* if $(a^{(n)})$ is Cauchy in ℓ^2, then for each k the numbers $a_k^{(n)}$ form a Cauchy sequence in $\mathbb{C}$, with limit a_k. For $\varepsilon > 0$ and $m, n \ge N_\varepsilon$ and every finite K: $\sum_{|k|\le K} |a_k^{(n)} - a_k^{(m)}|^2 < \varepsilon^2$; let $m \to \infty$, then $K \to \infty$: $\left\| a^{(n)} - a \right\| \le \varepsilon$. Hence $a \in \ell^2$ and $a^{(n)} \to a$.

(c) $L^2(X,\mu)$, see Theorem 1.13 for the notions involved:

$$L^2(X,\mu) := \Big\{ f\colon X \to \mathbb{C} \text{ measurable} : \int_X |f|^2\, d\mu < \infty \Big\} \Big/ \sim, \qquad (f,g) = \int_X f\, \overline{g}\, d\mu,$$

where $f \sim g$ if $f = g$ μ-almost everywhere on X. The quotient is needed for (ii) of Definition 1.2: $\int |f|^2 d\mu = 0$ only gives $f = 0$ a.e., not $f \equiv 0$. Completeness of L^2 is a theorem of measure theory (the *Riesz–Fischer theorem* in its original form).

Obviously $\ell^2(\mathbb{Z}) = L^2(\mathbb{Z}, \#)$, where $\#$ is the *counting measure* on $\mathbb{Z}$ ($\#(A)$ = number of elements of A; the notebook writes $\delta_{\mathbb{Z}}$ – the "discrete measure" $\sum_{n\in\mathbb{Z}} \delta_n$). Then $\int_{\mathbb{Z}} f\, d\# = \sum_{k\in\mathbb{Z}} f(k)$, and only the empty set is a null set.

(d) $S_2(H)$, the Hilbert–Schmidt operators on a Hilbert space H (Theorem 1.14), with $(A,B) = \operatorname{trace}(AB^*) = \sum_n (Ae_n, Be_n)$. For $H = \mathbb{C}^N$ these are all $N \times N$ matrices and $(A,B) = \sum_{i,j} A_{ij}\overline{B_{ij}}$ – the matrix is just regarded as a vector in $\mathbb{C}^{N^2}$.

(e) The *Fock space* (Bargmann–Fock space)

$$\mathcal{F}_2 = \Big\{ f \text{ entire} : \int_{\mathbb{C}} |f(z)|^2 e^{-|z|^2}\, d\lambda_2(z) < \infty \Big\}, \qquad (f,g) = \frac{1}{\pi} \int_{\mathbb{C}} f(z)\, \overline{g(z)}\, e^{-|z|^2}\, d\lambda_2(z),$$

where λ_2 is Lebesgue measure on $\mathbb{C} = \mathbb{R}^2$ and a function is *entire* if it is holomorphic (complex differentiable) on all of $\mathbb{C}$. See Example 1.15.

Background 13. (Measure spaces, measurable functions, a.e.)

A *σ-algebra* on a set X is a family $\mathcal{F}$ of subsets of X containing X and closed under complements and countable unions. A *measure* is a map $\mu\colon \mathcal{F} \to [0,\infty]$ with $\mu(\emptyset) = 0$ and $\mu(\bigsqcup_n A_n) = \sum_n \mu(A_n)$ for disjoint $A_n \in \mathcal{F}$. $(X, \mathcal{F}, \mu)$ is a *measure space*.
The *Borel σ-algebra* $\mathcal{B}(Y)$ of a metric (topological) space Y is the smallest σ-algebra containing all open sets. On $\mathbb{C}$ we use $\mathcal{B}(\mathbb{C}) = \mathcal{B}(\mathbb{R}^2)$. A measure on $\mathcal{B}(X)$ is a *Borel measure*.
$f\colon X \to \mathbb{C}$ is *measurable* if $f^{-1}(B) \in \mathcal{F}$ for every $B \in \mathcal{B}(\mathbb{C})$. A property holds *$\mu$-almost everywhere* (a.e.) if it fails only on a set $N \in \mathcal{F}$ with $\mu(N) = 0$.

Background 14. (Bounded operators, adjoints, Hilbert–Schmidt operators)

An *operator* $A\colon H \to H$ is a linear map. It is *bounded* if $\|A\| := \sup_{\|x\|\le 1} \|Ax\| < \infty$; $\mathcal{B}(H)$ denotes the bounded operators. For $A \in \mathcal{B}(H)$ the *adjoint* $A^* \in \mathcal{B}(H)$ is defined by $(Ax, y) = (x, A^*y)$ for all x, y (for matrices: the conjugate transpose). If $H = \ell^2$, an operator can be thought of as an infinite matrix $((Ae_j, e_i))_{i,j}$ acting on sequences.
$A \in \mathcal{B}(H)$ is *Hilbert–Schmidt* if $\sum_n \|Ae_n\|^2 < \infty$ for an orthonormal basis $\{e_n\}$ (Definition 1.19). The sum does not depend on the choice of the basis: by Parseval (Theorem 1.26) applied twice, for ONBs $\{e_n\}$, $\{f_m\}$,

$$\sum_n \|Ae_n\|^2 = \sum_n \sum_m |(Ae_n, f_m)|^2 = \sum_m \sum_n |(e_n, A^* f_m)|^2 = \sum_m \|A^* f_m\|^2,$$

and the right side does not depend on $\{e_n\}$. We set $\|A\|_{S_2} = \big(\sum_n \|Ae_n\|^2\big)^{1/2}$ and $S_2(H) = \{A \in \mathcal{B}(H) : \sum_n \|Ae_n\|^2 < \infty\}$. The *trace* of an operator is $\operatorname{trace} T = \sum_n (Te_n, e_n)$ (when this makes sense); for $A, B \in S_2(H)$ one has $\operatorname{trace}(B^*A) = \operatorname{trace}(AB^*) = \sum_n (Ae_n, Be_n)$, and this is the inner product of $S_2(H)$.

Example 1.15 (The Fock space)**.** Polynomials, constants, sin, e^{az}, … belong to $\mathcal{F}_2$. Indeed, in polar coordinates $z = re^{i\theta}$, $d\lambda_2 = r\, dr\, d\theta$,

$$\|z^n\|_{\mathcal{F}_2}^2 = \frac{1}{\pi} \int_0^{2\pi} \int_0^\infty r^{2n} e^{-r^2}\, r\, dr\, d\theta = 2 \int_0^\infty r^{2n+1} e^{-r^2} dr \overset{s=r^2}{=} \int_0^\infty s^n e^{-s} ds = n!,$$

and $(z^n, z^m) = 0$ for $n \neq m$ (the θ-integral of $e^{i(n-m)\theta}$ vanishes). So $\{z^n/\sqrt{n!}\}_{n\geq 0}$ is an orthonormal system in $\mathcal{F}_2$ (in fact an orthonormal basis). Since $|\sin z| \leq e^{|z|}$ and $|e^{az}| \leq e^{|a||z|}$, and $\int_{\mathbb{C}} e^{2c|z|-|z|^2} d\lambda_2 < \infty$, also sin and e^{az} belong to $\mathcal{F}_2$. On the other hand $e^{z^2} \notin \mathcal{F}_2$, since $|e^{z^2}|^2 e^{-|z|^2} = e^{|z|^2}$ on the real axis.

1.3 Orthonormal bases

Definition 1.16. Let H be a Hilbert space. A family $\{e_k\}_{k\in I} \subset H$ is an *orthogonal system* if $(e_k, e_j) = 0$ for all $k, j \in I$, $k \neq j$.

Background 17. (Span, closure, density)

$\operatorname{span}\{e_k\}_{k\in I}$ is the set of all *finite* linear combinations $\sum_{k\in F} c_k e_k$, $F \subset I$ finite. The *closure* $\overline{A} = \operatorname{clos}_H A$ of a set $A \subset H$ is the set of all limits of sequences in A; A is *dense* in H if $\overline{A} = H$, i.e. every $h \in H$ can be approximated arbitrarily well by elements of A: $\forall h \in H\ \forall \varepsilon > 0\ \exists a \in A : \|h - a\| < \varepsilon$. $\operatorname{clos}\operatorname{span}\{e_k\} = \overline{\operatorname{span}}\{e_k\}$ is the *closed linear span*: the smallest closed subspace containing all e_k.

Definition 1.18. Let H be a Hilbert space. A family $\{e_k\}_{k\in I}$ is *complete* if $\operatorname{clos}_H \operatorname{span}\{e_k\}_{k\in I} = H$.

Definition 1.19. $\{e_k\}_{k\in I}$ is an *orthonormal basis* (ONB) of H if it is a complete orthogonal system in H with $\|e_k\| = 1$ for all $k \in I$.

Remark 1.20. An ONB is usually *not* a basis in the sense of linear algebra (a Hamel basis): elements of H are in general *infinite* combinations $\sum_k c_k e_k$, converging in norm (Theorem 1.26). E.g. $(1, \frac{1}{2}, \frac{1}{3}, \dots) \in \ell^2(\mathbb{N})$ is not a finite combination of the unit vectors.

Background 21. (Separable spaces)

A metric space X is *separable* if it contains a *countable dense subset*, i.e. a countable $A \subseteq X$ with $\overline{A} = X$. Example: $\mathbb{R}$ is separable since $\mathbb{Q}$ is countable and $\overline{\mathbb{Q}} = \mathbb{R}$; $\mathbb{C}^N$ is separable ($(\mathbb{Q} + i\mathbb{Q})^N$). If A is countable and dense, then for every h there is a sequence $(a_n) \subset A$ with $a_n \to h$ (pick a_n with $\|h - a_n\| < \frac{1}{n}$).

Exercise 1.22. Let H be a separable Hilbert space and $\{e_k\}_{k\in I}$ an ONB in H. Then I is countable.

Solution. For $k \neq j$, by Pythagoras, $\|e_k - e_j\|^2 = \|e_k\|^2 + \|e_j\|^2 = 2$. Hence the open balls $B_k = B(e_k, \frac{\sqrt{2}}{2})$, $k \in I$, are pairwise disjoint (a point in $B_k \cap B_j$ would give $\|e_k - e_j\| < \sqrt{2}$). Let A be a countable dense set. Every ball B_k contains some point $a_k \in A$, and $a_k \neq a_j$ for $k \neq j$ by disjointness. Thus $k \mapsto a_k$ is an injection $I \to A$, so I is countable.

Proposition 1.23. *For a Hilbert space $H \neq \{0\}$ the following are equivalent: (a) H is separable; (b) H has a countable ONB; (c) H has a countable complete system.*

Proof. (a)⇒(c): a countable dense set is a complete system. (c)⇒(b): apply the Gram–Schmidt procedure to the complete sequence (x_n): discard every x_n that lies in $\operatorname{span}\{x_1, \dots, x_{n-1}\}$ and put $e_n = \frac{x_n - \sum_{j<n}(x_n, e_j)e_j}{\|\cdots\|}$. The e_n are orthonormal and have the same span as the x_n, hence they are complete. (b)⇒(a): finite combinations of the e_n with coefficients in $\mathbb{Q} + i\mathbb{Q}$ form a countable set whose closure contains $\operatorname{span}\{e_n\}$, hence equals H. □

From now on, all Hilbert spaces will be separable. To construct a non-separable Hilbert space one needs a "wild" measure, for example the counting measure on an uncountable set: $\ell^2(\mathbb{R}) = L^2(\mathbb{R}, \#)$ consists of $f : \mathbb{R} \to \mathbb{C}$ vanishing outside a countable set with $\sum_x |f(x)|^2 < \infty$, and $\{\mathbb{1}_{\{x\}}\}_{x\in\mathbb{R}}$ is an uncountable ONB.

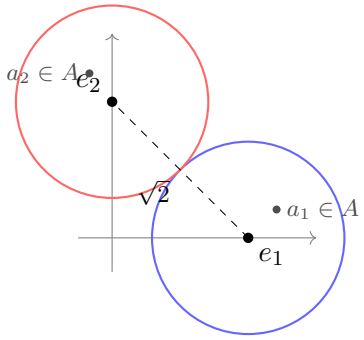


Figure 1: Orthonormal vectors are at distance $\sqrt{2}$, so balls of radius $\sqrt{2}/2$ around them are disjoint; each contains its own point of the dense set A.

1.4 Fourier coefficients and the Riesz–Fischer theorem

Example 1.24 (Observation: the finite-dimensional case)**.** In $\mathbb{C}^N$ the standard basis is $\{e_k\}_{k=1}^N$, $e_k = (0, \dots, 0, \underbrace{1}_{k\text{-th place}}, 0, \dots, 0)^T$, and for all $p \in \mathbb{C}^N$

$$p = p_1 e_1 + \cdots + p_N e_N, \qquad p_k = (p, e_k)_{\mathbb{C}^N}$$

(p_k is the projection coefficient of p onto e_k). This holds for *every* ONB $\{e_k\}$ of $\mathbb{C}^N$: if $p = \sum_j c_j e_j$, then $(p, e_k) = \sum_j c_j (e_j, e_k) = c_k$ by orthonormality. The goal is to prove a similar formula in the infinite-dimensional case.

Definition 1.25. Let H be a Hilbert space, $\{e_k\}_{k=1}^\infty$ an ONB in H and $h \in H$. The number

$$\hat{h}(k) = (h, e_k)$$

is the *k-th Fourier coefficient* of h with respect to $\{e_n\}_{n=1}^\infty$.

Theorem 1.26 (Riesz–Fischer)**.** *Let H be a separable Hilbert space with ONB $\{e_n\}_{n=1}^\infty$. Then for every $h \in H$*

$$h = \sum_{k=1}^{\infty} \hat{h}(k)\, e_k,$$

where the series converges in the norm of H, i.e. $\left\| h - \sum_{k=1}^N \hat{h}(k) e_k \right\|_H \xrightarrow{N\to\infty} 0$. Moreover

$$\|h\|^2 = \sum_{k=1}^{\infty} \left|\hat{h}(k)\right|^2 \qquad \textit{(Parseval's identity).} \tag{3}$$

Remark 1.27. (a) "Converges in the norm" is *not* pointwise convergence: for $H = L^2$ the partial sums need not converge at any given point (this will be the main theme of Section 2).

(b) Names vary in the literature: (3) is usually called *Parseval's identity*; "Riesz–Fischer theorem" often refers to completeness of L^2, or to the converse statement Corollary 1.30.

(c) The statement also holds for non-separable H (with uncountable sums, of which only countably many terms are non-zero), but we will not need this.

We need a simple but important fact first.

Lemma 1.28 (Continuity of the inner product)**.** *If $x_N \to x$ in H, then $(x_N, y) \to (x, y)$ for all $y \in H$, and $\|x_N\| \to \|x\|$.*

Proof. By Cauchy–Schwarz, $|(x_N, y) - (x, y)| = |(x_N - x, y)| \le \|x_N - x\| \, \|y\| \to 0$. By the triangle inequality, $\|x\| \le \|x_N\| + \|x - x_N\|$ and $\|x_N\| \le \|x\| + \|x_N - x\|$, hence $\big|\|x_N\| - \|x\|\big| \le \|x_N - x\| \to 0$. □

Proof of Theorem 1.26. Step 1: Bessel's inequality. We show

$$\sum_{k=1}^{\infty} \left|\hat{h}(k)\right|^2 \le \|h\|^2 . \tag{4}$$

For this, completeness of $\{e_k\}$ is not needed – it holds for every orthonormal system. Set $h_N = \sum_{k=1}^{N} \hat{h}(k) e_k$. By orthonormality,

$$\Big\|\sum_{k=1}^{N} c_k e_k\Big\|^2 = \Big(\sum_k c_k e_k, \sum_j c_j e_j\Big) = \sum_{k,j} c_k \overline{c_j}\,(e_k, e_j) = \sum_{k=1}^{N} |c_k|^2, \qquad \text{so} \qquad \|h_N\|^2 = \sum_{k=1}^{N} \left|\hat{h}(k)\right|^2 .$$

Further, by Remark 1.3,

$$(h, h_N) = \Big(h, \sum_{k=1}^{N} \hat{h}(k) e_k\Big) = \sum_{k=1}^{N} \overline{\hat{h}(k)}\,(h, e_k) = \sum_{k=1}^{N} \overline{\hat{h}(k)} \hat{h}(k) = \sum_{k=1}^{N} \left|\hat{h}(k)\right|^2 = \|h_N\|^2 .$$

Summarizing, $(h, h_N) = \|h_N\|^2$, so by Cauchy–Schwarz $\|h_N\|^2 \le \|h\| \, \|h_N\|$, i.e. $\|h_N\| \le \|h\|$ (trivial if $h_N = 0$). Thus $\sum_{k=1}^{N} \left|\hat{h}(k)\right|^2 \le \|h\|^2$ for all N, and letting $N \to \infty$ gives (4).

Step 2: (h_N) is a Cauchy sequence. For $N > M$,

$$\|h_N - h_M\|^2 = \Big\| \sum_{k=M+1}^{N} \hat{h}(k) e_k \Big\|^2 = \sum_{k=M+1}^{N} \left|\hat{h}(k)\right|^2 \le \sum_{k=M+1}^{\infty} \left|\hat{h}(k)\right|^2 \xrightarrow{M\to\infty} 0,$$

since the tails of the convergent series (4) tend to 0. As H is complete, there exists

$$\tilde{h} = \lim_{N\to\infty} h_N = \lim_{N\to\infty} \sum_{k=1}^{N} \hat{h}(k) e_k,$$

in other words $\sum_{k=1}^{\infty} \hat{h}(k) e_k$ converges and $\tilde{h} = \sum_{k=1}^{\infty} \hat{h}(k) e_k$.

Step 3: $\tilde{h} = h$.

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For fixed k and all $N \ge k$ we have $(h_N, e_k) = \sum_{j=1}^{N} \hat{h}(j)\,(e_j, e_k) = \hat{h}(k)$. Hence, by Lemma 1.28 (this is the step that was "left as an exercise"),

$$\big(\tilde{h}, e_k\big) = \big(\lim_{N\to\infty} h_N, e_k\big) = \lim_{N\to\infty} (h_N, e_k) = \hat{h}(k) = (h, e_k)\,,$$

so $\big(h - \tilde{h}, e_k\big) = 0$ for all k: h and $\tilde{h}$ have the same projections onto every e_k. By linearity $\big(h - \tilde{h}, g\big) = 0$ for all $g \in \operatorname{span}\{e_k\}$, and by Lemma 1.28 (in the second slot) for all $g \in \operatorname{clos}\operatorname{span}\{e_k\} = H$ – here completeness of the system is used. Taking $g := h - \tilde{h}$ gives $\big\|h - \tilde{h}\big\|^2 = 0$, i.e. $h = \tilde{h}$.

Step 4: Parseval. Since $h_N \to h$, Lemma 1.28 gives

$$\|h\|^2 = \lim_{N\to\infty} \|h_N\|^2 = \lim_{N\to\infty} \sum_{k=1}^{N} \left|\hat{h}(k)\right|^2 = \sum_{k=1}^{\infty} \left|\hat{h}(k)\right|^2 .$$

□

Corollary 1.29 (Completeness criterion)**.** *An orthonormal sequence $\{e_k\}$ in H is complete if and only if the only $h \in H$ with $(h, e_k) = 0$ for all k is $h = 0$.*

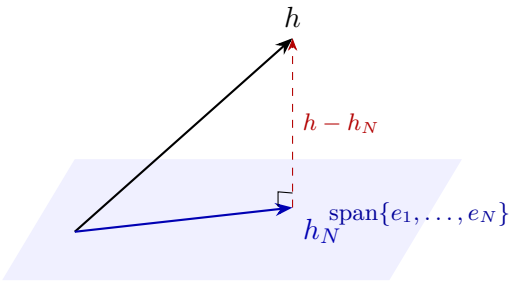


Figure 2: h_N is the orthogonal projection of h onto $\operatorname{span}\{e_1, \dots, e_N\}$: $h - h_N \perp e_k$ for $k \le N$. Pythagoras gives $\|h\|^2 = \|h_N\|^2 + \|h - h_N\|^2 \ge \|h_N\|^2$ – another proof of Bessel's inequality.

Proof. "$\Rightarrow$" is Step 3 above. "$\Leftarrow$": for any h, Steps 1–2 (which do not use completeness) give $\tilde{h} = \sum \hat{h}(k) e_k \in \operatorname{clos}\operatorname{span}\{e_k\}$, and as in Step 3, $\left(h - \tilde{h}, e_k\right) = 0$ for all k. By assumption $h = \tilde{h} \in \operatorname{clos}\operatorname{span}\{e_k\}$. □

Corollary 1.30. *If $\{e_k\}$ is an ONB of H and $(c_k) \in \ell^2(\mathbb{N})$, then $\sum_k c_k e_k$ converges to some $h \in H$ with $\hat{h}(k) = c_k$. Consequently $h \mapsto (\hat{h}(k))_k$ is a bijective linear isometry $H \to \ell^2(\mathbb{N})$: every separable infinite-dimensional Hilbert space "is" ℓ^2.*

Proof. The partial sums are Cauchy exactly as in Step 2; the limit has coefficients c_k as in Step 3. □

Example 1.31 (Euler's formula)**.** $\displaystyle\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}.$

Proof. Let $H = L^2([-\pi, \pi], \mu = \frac{dx}{2\pi})$ and $e_k(x) = e^{ikx}$, $k \in \mathbb{Z}$. Orthonormality is a direct computation: $(e_k, e_j) = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{i(k-j)x} dx = 1$ if $k = j$ and $= \frac{1}{2\pi}\frac{e^{i(k-j)x}}{i(k-j)}\Big|_{-\pi}^{\pi} = 0$ if $k \ne j$. The non-trivial thing is that $\{e_k\}_{k\in\mathbb{Z}}$ is *complete*: "trigonometric functions are dense in L^2". This is proved in Corollary 2.19 below (and nothing proved before it relies on this example). Since $\mathbb{Z}$ is countable, we may enumerate $\{e_k\}_{k\in\mathbb{Z}}$ as a sequence and apply Theorem 1.26.

Consider $f(x) = x$ on $[-\pi, \pi]$. Then $\|f\|_H^2 = \frac{1}{2\pi}\int_{-\pi}^{\pi} x^2 dx = \frac{1}{2\pi}\cdot\frac{2\pi^3}{3} = \frac{\pi^2}{3}$, $\hat{f}(0) = \frac{1}{2\pi}\int_{-\pi}^{\pi} x\, dx = 0$, and for $k \ne 0$, integrating by parts,

$$\hat{f}(k) = \frac{1}{2\pi}\int_{-\pi}^{\pi} x\,\overline{e^{ikx}}\,dx = \frac{1}{2\pi}\int_{-\pi}^{\pi} x e^{-ikx} dx = \frac{1}{2\pi}\Big[\frac{xe^{-ikx}}{-ik}\Big]_{-\pi}^{\pi} + \frac{1}{2\pi}\underbrace{\int_{-\pi}^{\pi}\frac{e^{-ikx}}{ik}dx}_{=0}$$

$$= \frac{1}{2\pi}\cdot\frac{\pi e^{-ik\pi} + \pi e^{ik\pi}}{-ik} = \frac{(-1)^k}{-ik} = \frac{i(-1)^k}{k}.$$

Hence $\sum_{k\in\mathbb{Z}}\left|\hat{f}(k)\right|^2 = \sum_{k\ne 0}\frac{1}{k^2} = 2\sum_{k=1}^{\infty}\frac{1}{k^2}$, and Parseval's identity $\|f\|_H^2 = \sum_k |\hat{f}(k)|^2$ – "Pythagoras in infinite dimensions" – gives $\frac{\pi^2}{3} = 2\sum_{k\ge 1}\frac{1}{k^2}$. □

2 Classical Fourier series

2.1 Setting

Let $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$ be the unit circle and m the Lebesgue (arc-length) measure on $\mathbb{T}$ normalized by $\frac{1}{2\pi}$, so that $m(\mathbb{T}) = 1$. Concretely, $t \mapsto e^{it}$ maps $[-\pi, \pi)$ bijectively onto $\mathbb{T}$ and

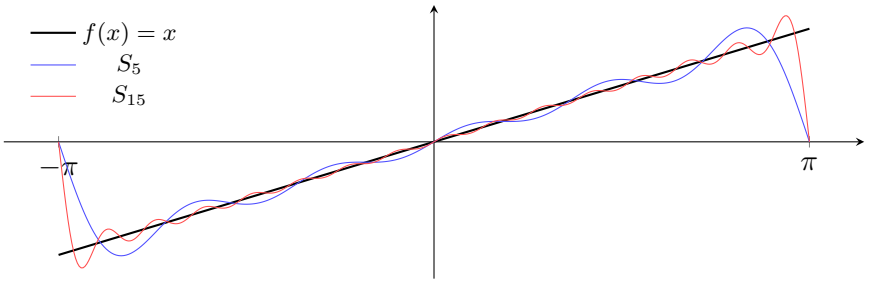


Figure 3: Partial sums $S_N(x) = \sum_{|k|\le N} \hat{f}(k) e^{ikx} = 2\sum_{k=1}^{N} \frac{(-1)^{k+1}}{k} \sin kx$ of $f(x) = x$. They converge to f in L^2; at $\pm\pi$ (the jump of the periodic extension) they are 0, the average of the one-sided limits (cf. Theorem 2.36).

$m(A) = \frac{1}{2\pi}\lambda_1\{t \in [-\pi, \pi) : e^{it} \in A\}$. We work in

$$H = L^2(\mathbb{T}, m), \qquad (f, g) = \underbrace{\int_{\mathbb{T}} f\,\overline{g}\,dm}_{\text{general measure theory}} = \underbrace{\frac{1}{2\pi}\int_{-\pi}^{\pi} f(e^{it})\,\overline{g(e^{it})}\,dt}_{\text{parametrization}}.$$

Functions on $\mathbb{T}$ are the same as 2π-periodic functions on $\mathbb{R}$, via $g(t) = f(e^{it})$.

Exercise 2.1. $\{z^n\}_{n\in\mathbb{Z}}$ is an ONB in $L^2(\mathbb{T}, m)$.

Solution. Orthonormality: $\left(z^n, z^k\right) = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{i(n-k)t}dt = \delta_{nk}$, as in Example 1.31. *Completeness* needs two facts.

(1) Trigonometric polynomials $\sum_{|k|\le N} c_k z^k$ are dense in $C(\mathbb{T})$ with the sup-norm. This follows from the Stone–Weierstrass theorem (Theorem 2.2), since they form an algebra containing 1, closed under conjugation ($\overline{z} = z^{-1}$ on $\mathbb{T}$) and separating points (already z does). We will also get a self-contained proof from Fejér's theorem (Corollary 2.18).

(2) $C(\mathbb{T})$ is dense in $L^2(\mathbb{T}, m)$ (Theorem 2.3).

Given $f \in L^2$ and $\varepsilon > 0$, choose $g \in C(\mathbb{T})$ with $\|f - g\|_{L^2} < \varepsilon$ and a trigonometric polynomial p with $\max_{\mathbb{T}} |g - p| < \varepsilon$. Since $m(\mathbb{T}) = 1$, $\|g - p\|_{L^2} \le \max_{\mathbb{T}} |g - p|$, so $\|f - p\|_{L^2} < 2\varepsilon$. Hence $\operatorname{clos}\operatorname{span}\{z^n\} = L^2(\mathbb{T}, m)$.

Background 2. (Weierstrass and Stone–Weierstrass theorems)

Weierstrass approximation theorem. Every continuous $f\colon [a, b] \to \mathbb{C}$ is a uniform limit of polynomials. *Trigonometric version:* every $f \in C(\mathbb{T})$ is a uniform limit of trigonometric polynomials.
Stone–Weierstrass theorem. Let K be a compact metric space and $\mathcal{A} \subset C(K)$ a subalgebra (closed under sums, products and scalar multiples) which contains the constants, separates points ($\forall x \ne y\ \exists a \in \mathcal{A} : a(x) \ne a(y)$) and is closed under complex conjugation. Then $\mathcal{A}$ is dense in $C(K)$ with respect to $\|f\|_\infty = \max_K |f|$.

Background 3. (Continuous functions are dense in L^p)

If μ is a finite Borel measure on a compact metric space K and $1 \le p < \infty$, then $C(K)$ is dense in $L^p(K, \mu)$. (Idea: simple functions are dense; by regularity of μ an indicator $\mathbb{1}_A$ is approximated by $\mathbb{1}_F$ with $F \subset A$ closed, and $\mathbb{1}_F$ by the continuous functions $\max(0, 1 - n\operatorname{dist}(x, F))$.)

Definition 2.4. For $f \in L^2(\mathbb{T}, m)$ (or even $f \in L^1(\mathbb{T}, m)$) the Fourier coefficients are

$$\hat{f}(k) = \left(f, z^k\right) = \int_{\mathbb{T}} f(z)\,\overline{z}^k\,dm(z) = \int_{\mathbb{T}} f(z)\,z^{-k}\,dm(z) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(e^{it})e^{-ikt}dt, \qquad k \in \mathbb{Z},$$

using $\overline{z} = z^{-1}$ on $\mathbb{T}$. The *Fourier series* of f is $\sum_{k\in\mathbb{Z}} \hat{f}(k) z^k$ and its symmetric partial sums are $S_N(f,z) = \sum_{k=-N}^{N} \hat{f}(k) z^k$.

By the Riesz–Fischer theorem, for all $f \in L^2(\mathbb{T}, m)$ we have $f = \sum_{k\in\mathbb{Z}} \hat{f}(k) z^k$ where the series converges in $L^2(\mathbb{T}, m)$, i.e.

$$\int_{\mathbb{T}} \Big| f - \sum_{k=-N}^{N} \hat{f}(k) z^k \Big|^2 dm \xrightarrow{N\to\infty} 0. \tag{$**$}$$

(Here we use that $\mathbb{Z}$ can be enumerated as $0, 1, -1, 2, -2, \dots$, so the symmetric partial sums are a subsequence of the partial sums of Theorem 1.26.)

But ($**$) is not what one would naively expect: we are interested in *pointwise* convergence. In other words, we would like to have $f(z) = \sum_{k\in\mathbb{Z}} \hat{f}(k) z^k$ a.e. on $\mathbb{T}$ for "good" f. This is true, but very hard to prove:

Theorem 2.5 (Carleson 1966, Hunt 1968)**.** *If $f \in L^p(\mathbb{T}, m)$ for some $p \in (1, \infty]$, then $S_N(f,z) \xrightarrow{N\to\infty} f(z)$ for m-almost every $z \in \mathbb{T}$.*

For $p = 1$ this fails: Kolmogorov (1923) constructed $f \in L^1(\mathbb{T})$ whose Fourier series diverges almost everywhere (even everywhere). We will prove much simpler results, starting with convergence in a weaker sense.

Definition 2.6 (Cesàro convergence)**.** A sequence $(S_n)_{n\geq 0}$ *converges in the Cesàro sense* to L if its arithmetic means converge: $\sigma_N := \frac{1}{N+1} \sum_{n=0}^{N} S_n \xrightarrow{N\to\infty} L$. A series converges in the Cesàro sense if the sequence of its partial sums does.

Example 2.7. $S_n = -1, 1, -1, 1, \dots$ (the partial sums of $-1 + 2 - 2 + 2 - \dots$) does not converge, but $\sigma_N \to 0$: the means are 0 or $-\frac{1}{N+1}$.

Proposition 2.8 (Cesàro summation is consistent)**.** *If $S_n \to L$, then also $\sigma_N \to L$.*

Proof. Let $\varepsilon > 0$ and $|S_n - L| < \varepsilon$ for $n > n_0$. Then $|\sigma_N - L| \leq \frac{1}{N+1} \sum_{n=0}^{n_0} |S_n - L| + \frac{N-n_0}{N+1}\varepsilon$, and the first term tends to 0 as $N \to \infty$. So $\limsup |\sigma_N - L| \leq \varepsilon$ for every ε. □

Theorem 2.9 (Fejér)**.** *Let $f \in C(\mathbb{T})$ (a continuous function on the unit circle). Then the Fourier series $\sum_{k=-\infty}^{\infty} \hat{f}(k) z^k$ converges to f in the Cesàro sense everywhere on $\mathbb{T}$. In other words, if $S_N(z) = \sum_{k=-N}^{N} \hat{f}(k) z^k$, then*

$$\frac{1}{N+1} \sum_{n=0}^{N} S_n(z) \longrightarrow f(z) \qquad \text{for every } z \in \mathbb{T}.$$

The proof is given in Section 2.4; first we write the Cesàro means as an integral.

2.2 The Fejér kernel

Lemma 2.10. $\displaystyle \frac{1}{N+1} \sum_{n=0}^{N} S_n(z) = \int_{\mathbb{T}} f(\xi)\, F_N(\xi, z)\, dm(\xi)$, *where*

$$F_N(\xi, z) := \frac{|1 + \overline{\xi} z + \cdots + (\overline{\xi} z)^N|^2}{N+1} = \frac{1}{N+1} \Big| \sum_{s=0}^{N} (\overline{\xi} z)^s \Big|^2$$

is the Fejér kernel.

Proof. Since all sums are finite, we may interchange them with the integral:

$$\frac{1}{N+1}\sum_{n=0}^{N} S_n(z) = \frac{1}{N+1}\sum_{n=0}^{N}\sum_{k=-n}^{n} \hat f(k) z^k = \frac{1}{N+1}\sum_{n=0}^{N}\sum_{k=-n}^{n}\int_{\mathbb{T}} f(\xi)\overline{\xi}^k \, dm(\xi)\, z^k$$
$$= \int_{\mathbb{T}} f(\xi)\Big(\frac{1}{N+1}\sum_{n=0}^{N}\sum_{k=-n}^{n} \overline{\xi}^k z^k\Big) dm(\xi).$$

So the lemma reduces to the identity, for $w = \overline{\xi} z \in \mathbb{T}$,

$$\sum_{n=0}^{N}\sum_{k=-n}^{n} w^k = \Big|\sum_{s=0}^{N} w^s\Big|^2. \tag{5}$$

Direct proof. Using $\overline{w} = w^{-1}$,

$$\Big|\sum_{s=0}^{N} w^s\Big|^2 = \Big(\sum_{s=0}^{N} w^s\Big)\Big(\sum_{r=0}^{N} w^{-r}\Big) = \sum_{k=-N}^{N} \#\{(s,r) : s-r=k\}\, w^k = \sum_{k=-N}^{N} (N+1-|k|)\, w^k,$$

and on the left of (5) the power w^k appears once for every n with $|k| \le n \le N$, i.e. $N+1-|k|$ times.

Proof by induction (as in the lecture). For $N=0$ both sides equal 1. Write $P_N = \sum_{s=0}^{N} w^s$ and suppose (5) holds for N. The left side for $N+1$ is $\sum_{k=-(N+1)}^{N+1} w^k + |P_N|^2$. Using $|a+b|^2 = |a|^2 + |b|^2 + 2\,\mathrm{Re}(a\bar b)$ with $a = P_N$, $b = w^{N+1}$:

$$|P_{N+1}|^2 = |P_N|^2 + \underbrace{|w^{N+1}|^2}_{=1} + 2\,\mathrm{Re}\big(P_N \overline{w}^{N+1}\big), \qquad P_N\overline{w}^{N+1} = \sum_{s=0}^{N} w^{s-N-1} = \overline{w} + \overline{w}^2 + \cdots + \overline{w}^{N+1},$$

so $|P_{N+1}|^2 = |P_N|^2 + 1 + \sum_{j=1}^{N+1}(w^j + \overline{w}^j) = |P_N|^2 + \sum_{k=-(N+1)}^{N+1} w^k$, as needed. □

Remark 2.11 (Closed form; Dirichlet kernel). Write $z = e^{i\theta}$, $\xi = e^{it}$, $u = \theta - t$, so $w = \overline{\xi} z = e^{iu}$. Summing the geometric series and using $|1 - e^{ia}| = 2|\sin\frac{a}{2}|$,

$$F_N(\xi, z) = K_N(u) := \frac{1}{N+1}\frac{|1-e^{i(N+1)u}|^2}{|1-e^{iu}|^2} = \frac{1}{N+1}\left(\frac{\sin\frac{(N+1)u}{2}}{\sin\frac{u}{2}}\right)^2.$$

Similarly $S_N(f, e^{i\theta}) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(e^{it}) D_N(\theta - t)\, dt$ with the *Dirichlet kernel*

$$D_N(u) = \sum_{k=-N}^{N} e^{iku} = \frac{\sin\big((N+\frac12)u\big)}{\sin\frac{u}{2}}, \qquad \text{and} \qquad K_N = \frac{1}{N+1}\sum_{n=0}^{N} D_n.$$

The decisive difference: $K_N \ge 0$ while D_N changes sign, and $\frac{1}{2\pi}\int |D_N| \to \infty$ (Section 2.5). Both kernels are even functions of u.

2.3 Approximate identities

Definition 2.12 (Approximate identity)**.** Let X be a metric space with metric ρ, μ a Borel measure on X and $x_0 \in X$. An *approximate identity with center at* x_0 is a family of measurable functions p_n on X such that

(i) $p_n(x) \ge 0$ for all $x \in X$;

(ii) $\int_X p_n(x)\, d\mu(x) = 1$ for all n;

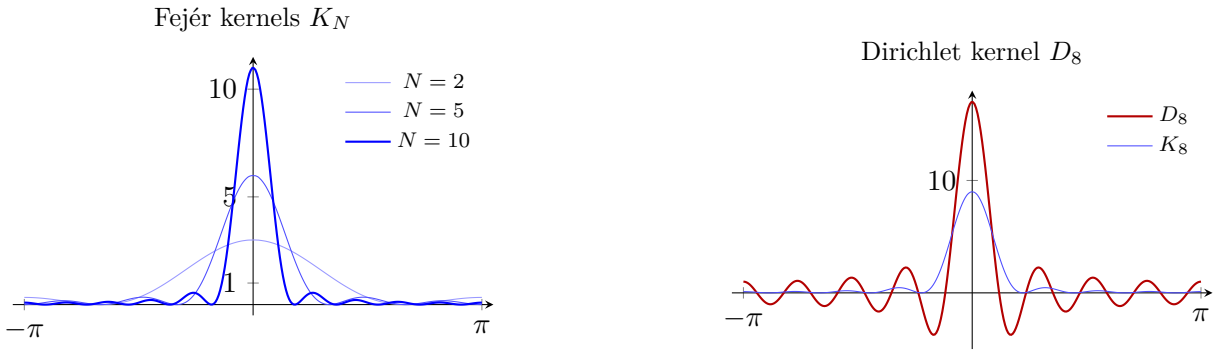


Figure 4: Left: $K_N \geq 0$, $\frac{1}{2\pi}\int K_N = 1$, and the mass concentrates at 0 – an approximate identity. Right: the Dirichlet kernel oscillates and takes negative values; its absolute integral grows like $\log N$.

(iii) $\int_{X\setminus B(x_0,\delta)} p_n(x)\,d\mu(x) \xrightarrow{n\to\infty} 0$ for every $\delta > 0$, where $B(x_0,\delta) = \{y \in X : \rho(y,x_0) < \delta\}$.

Remark 2.13. (a) In the lecture (iii) was first stated as $\sup_{x\in X\setminus B(x_0,\delta)} p_n(x) \to 0$; the integral version above (suggested in the lecture as "simpler") is weaker when $\mu(X) < \infty$, since $\int_{X\setminus B} p_n\,d\mu \leq \mu(X)\sup_{X\setminus B} p_n$, and it is exactly what the proof below needs. For the Fejér kernel both hold.

(b) Whether the ball is open or closed does not matter.

(c) Intuition: the mass 1 of $p_n\,d\mu$ concentrates at x_0. In the language of distributions, $p_n \to \delta_{x_0}$ (Dirac delta): $\int f\,p_n\,d\mu \to \int f\,d\delta_{x_0} = f(x_0)$ – that is the next lemma.

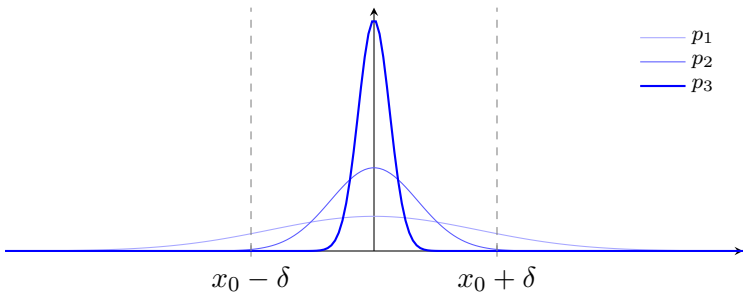


Figure 5: An approximate identity: non-negative, total mass 1, and the mass outside $B(x_0,\delta)$ tends to 0.

Lemma 2.14. *Let (p_n) be an approximate identity with center at x_0 and let f be a bounded measurable function on X which is continuous at x_0 (e.g. f bounded and continuous). Then*

$$\int_X f(x)\,p_n(x)\,d\mu(x) \xrightarrow{n\to\infty} f(x_0).$$

Proof. By (ii), $f(x_0) = \int_X f(x_0)p_n\,d\mu$, so using (i),

$$\begin{aligned}\Big|\int_X f\,p_n\,d\mu - f(x_0)\Big| &= \Big|\int_X (f(x)-f(x_0))p_n(x)\,d\mu(x)\Big| \\ &\leq \int_{B(x_0,\delta)} |f(x)-f(x_0)|\,p_n(x)\,d\mu + \int_{X\setminus B(x_0,\delta)} |f(x)-f(x_0)|\,p_n(x)\,d\mu \\ &\leq \sup_{x\in B(x_0,\delta)} |f(x)-f(x_0)| \cdot \underbrace{\int_{B(x_0,\delta)} p_n\,d\mu}_{\leq 1 \text{ by (i),(ii)}} + 2\sup_{y\in X}|f(y)| \cdot \int_{X\setminus B(x_0,\delta)} p_n\,d\mu.\end{aligned}$$

Let $\varepsilon > 0$. By continuity at x_0 choose $\delta > 0$ with $|f(x) - f(x_0)| < \frac{\varepsilon}{2}$ on $B(x_0, \delta)$; then, by (iii), choose N such that the second summand is $< \frac{\varepsilon}{2}$ for $n \geq N$. □

We will see that $F_N(\xi, z)$ is an approximate identity (in ξ) at the point z.

Remark 2.15 (Dense subsets, once more). "A subset $D = \{x_1, x_2, x_3, \dots\}$ is dense in a Hilbert space H if for every $h \in H$ and every $\varepsilon > 0$ there exists some $x_n \in D$ such that $\|h - x_n\|_H < \varepsilon$." This is the definition from Theorem 1.17 specialised to countable sets; such H is separable (Theorem 1.21).

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2.4 Proof of Fejér's theorem

Recap. $H = L^2(\mathbb{T}, m)$, $\mathbb{T} = \{|z| = 1\}$, ONB $\{z^k\}_{k \in \mathbb{Z}}$,

$$S_n(f, z) := \sum_{k=-n}^{n} \hat{f}(k) z^k, \qquad \sigma_N(f, z) := \frac{1}{N+1} \sum_{n=0}^{N} S_n(f, z).$$

Theorem 2.16 (Fejér)**.** *For every $f \in C(\mathbb{T})$ we have $\sigma_N(f, z) \xrightarrow{N \to \infty} f(z)$ for every $z \in \mathbb{T}$. Equivalently, the Fourier series $\sum_{k=-\infty}^{\infty} \hat{f}(k) z^k$ converges to $f(z)$ in the Cesàro sense at every point $z \in \mathbb{T}$.*

Proof. By Lemma 2.10, $\sigma_N(f, z) = \int_{\mathbb{T}} f(\xi) F_N(\xi, z)\, dm(\xi)$ with $F_N(\xi, z) = \frac{1}{N+1} \left|\sum_{s=0}^{N} (\overline{\xi} z)^s\right|^2$. Fix z. We check that $(F_N(\cdot, z))_N$ is an approximate identity in the variable ξ with center at z, on $X = \mathbb{T}$ with $\rho(\xi, z) = |\xi - z|$ and $\mu = m$:

(i) $F_N(\xi, z) \geq 0$ for all $\xi \in \mathbb{T}$. ✓

(ii) $\int_{\mathbb{T}} F_N(\xi, z)\, dm(\xi) = 1$. Indeed, $\xi \mapsto 1 + \overline{\xi} z + \dots + \overline{\xi}^N z^N = \sum_{s=0}^{N} z^s\, \xi^{-s}$ is a finite combination of the orthonormal functions ξ^{-s}, so by Parseval (or just orthonormality)

$$\int_{\mathbb{T}} \left|1 + \overline{\xi} z + \dots + \overline{\xi}^N z^N\right|^2 dm(\xi) = 1 + |z|^2 + |z^2|^2 + \dots + |z^N|^2 \overset{|z|=1}{=} N + 1. ✓$$

(iii) For $\xi \in \mathbb{T} \setminus B(z, \delta)$ we have $|1 - \overline{\xi} z| = |\overline{\xi}|\, |\xi - z| = |\xi - z| \geq \delta$, and $|1 - \overline{\xi}^{N+1} z^{N+1}| \leq 2$. Summing the geometric series,

$$\int_{\mathbb{T} \setminus B(z,\delta)} F_N(\xi, z)\, dm(\xi) \leq \underbrace{m(\mathbb{T} \setminus B(z, \delta))}_{\leq m(\mathbb{T}) = 1} \cdot \sup_{\xi \in \mathbb{T} \setminus B(z,\delta)} \frac{1}{N+1} \left|\frac{1 - \overline{\xi}^{N+1} z^{N+1}}{1 - \overline{\xi} z}\right|^2$$
$$\leq \frac{2^2}{\delta^2 (N+1)} \xrightarrow[\delta \text{ fixed}]{N \to \infty} 0.$$

So $F_N(\cdot, z)$ is an approximate identity with center at z, and f is bounded and continuous on the compact set $\mathbb{T}$. By Lemma 2.14, $\sigma_N(f, z) \to f(z)$, for every z. □

Remark 2.17 (Uniform convergence). In fact $\sigma_N(f, \cdot) \rightrightarrows f$ uniformly on $\mathbb{T}$, i.e. $\max_{z \in \mathbb{T}} |\sigma_N(f, z) - f(z)| \to 0$, $N \to \infty$, because all estimates were uniform in z. *Check:* f is uniformly continuous on the compact set $\mathbb{T}$, so for $\varepsilon > 0$ there is δ *not depending on* z with $|f(\xi) - f(z)| < \frac{\varepsilon}{2}$ whenever $|\xi - z| < \delta$. The proof of Lemma 2.14 then gives

$$|\sigma_N(f, z) - f(z)| \leq \frac{\varepsilon}{2} + 2\, \|f\|_\infty \cdot \frac{4}{\delta^2 (N+1)} \qquad \text{for all } z \in \mathbb{T},$$

and the right side is $< \varepsilon$ for N large, independently of z.

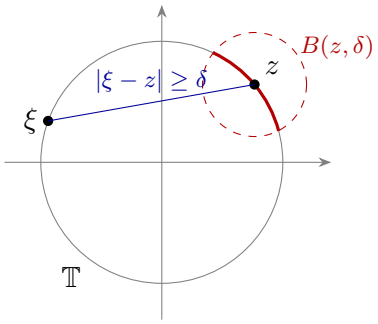


Figure 6: For ξ outside the arc $B(z,\delta)\cap\mathbb{T}$ the denominator $|1-\overline{\xi}z| = |\xi - z|$ stays $\geq \delta$, so $F_N(\xi,z) \leq \frac{4}{\delta^2(N+1)}$ there.

Corollary 2.18 (Weierstrass' trigonometric approximation theorem)**.** *Trigonometric polynomials are dense in* $(C(\mathbb{T}), \|\cdot\|_\infty)$.

Proof. $\sigma_N(f,\cdot) = \sum_{|k|\leq N}(1-\frac{|k|}{N+1})\hat{f}(k)z^k$ is a trigonometric polynomial and $\sigma_N(f,\cdot)\to f$ uniformly by Remark 2.17. □

Corollary 2.19. $\{z^n\}_{n\in\mathbb{Z}}$ *is an ONB of* $L^2(\mathbb{T},m)$*;*
equivalently $\{e^{ikx}\}_{k\in\mathbb{Z}}$ *is an ONB of* $L^2([-\pi,\pi],\frac{dx}{2\pi})$*. In particular Example 1.31 is justified.*

Proof. As in the solution of Exercise 2.1, with Corollary 2.18 in place of Stone–Weierstrass. □

Corollary 2.20. *If* $f\in C(\mathbb{T})$ *and* $S_N(f,z)$ *converges at some point* z*, then the limit is* $f(z)$ *(by Proposition 2.8 and Fejér's theorem).*

Remark 2.21 (Continuity at one point suffices). If f is bounded and measurable on $\mathbb{T}$ (not necessarily in $C(\mathbb{T})$) and $f(\xi)\to f(z_0)$ as $\xi\to z_0$, then $\sigma_N(f,z_0)\to f(z_0)$. *Check:* Lemma 2.14 only uses boundedness of f and continuity at the center $x_0 = z_0$.

Remark 2.22 (Jump discontinuities – homework). Let f be bounded and measurable, and let $z_0 = e^{it_0}$ be such that the one-sided limits

$$f(z_0-0) := \lim_{\varepsilon\to 0,\ \varepsilon>0} f(e^{i(t_0-\varepsilon)}), \qquad f(z_0+0) := \lim_{\varepsilon\to 0,\ \varepsilon>0} f(e^{i(t_0+\varepsilon)})$$

exist (f need not be continuous at z_0). Then

$$\sigma_N(f,z_0) \longrightarrow \frac{f(z_0-0)+f(z_0+0)}{2} \qquad (N\to\infty).$$

This generalizes Fejér's theorem (if f is continuous at z_0 both limits equal $f(z_0)$).

Solution. By Remark 2.11, $\sigma_N(f,z_0) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(e^{i(t_0+u)})K_N(u)\,du$ (substitute $t = t_0+u$ and use that K_N is even and 2π-periodic). Since K_N is even and $\frac{1}{2\pi}\int_{-\pi}^{\pi}K_N = 1$, we have $\frac{1}{2\pi}\int_0^{\pi}K_N = \frac{1}{2\pi}\int_{-\pi}^{0}K_N = \frac{1}{2}$. Hence, with $L_\pm = f(z_0\pm 0)$,

$$\sigma_N(f,z_0) - \frac{L_-+L_+}{2} = \frac{1}{2\pi}\int_0^{\pi}(f(e^{i(t_0+u)})-L_+)K_N(u)\,du + \frac{1}{2\pi}\int_0^{\pi}(f(e^{i(t_0-u)})-L_-)K_N(u)\,du.$$

Each term tends to 0 by the argument of Lemma 2.14: given ε, choose δ such that $|f(e^{i(t_0+u)})-L_+|<\varepsilon$ for $0<u<\delta$; the part $\int_0^\delta$ is $\leq\frac{\varepsilon}{2}$, and the part $\int_\delta^\pi$ is $\leq 2\|f\|_\infty\cdot\frac{1}{2\pi}\int_{\delta\leq|u|\leq\pi}K_N\to 0$, because $K_N(u)\leq\frac{1}{(N+1)\sin^2(\delta/2)}$ there.

2.5 Divergence: the du Bois-Reymond theorem

Theorem 2.23 (du Bois-Reymond)**.** *There exists a function $f \in C(\mathbb{T})$ such that $S_N(f,z)$ does not converge at some point $z \in \mathbb{T}$.*

So for continuous functions the Cesàro means in Fejér's theorem are really necessary. The proof is based on a classical result of functional analysis.

Background 24. (Banach spaces, bounded operators and functionals)

A *Banach space* is a complete normed linear space (Theorem 1.9). Example: $C(\mathbb{T})$ with $\|f\| = \max_{z\in\mathbb{T}} |f(z)|$ – a uniform limit of continuous functions is continuous, so Cauchy sequences converge in $C(\mathbb{T})$. A linear map $T\colon X \to Y$ between normed spaces is *bounded* if $\|T\| := \sup_{\|x\|\le 1} \|Tx\| < \infty$; for linear maps, bounded $\iff$ continuous. A bounded linear map $X \to \mathbb{C}$ is a *bounded linear functional*.

Theorem 2.25 (Banach–Steinhaus, uniform boundedness principle)**.** *Let X be a Banach space, Y a normed linear space and $\{T_\alpha\}_{\alpha\in A}$ a family of bounded linear operators $X \to Y$ such that $\sup_{\alpha\in A} \|T_\alpha x\| < \infty$ for every $x \in X$. Then*

$$\sup_{\alpha\in A} \|T_\alpha\| = \sup_{\alpha\in A} \sup_{x\in X,\ \|x\|\le 1} \|T_\alpha x\| < \infty.$$

The proof was skipped in the lecture as a basic fact from functional analysis; here is the standard argument.

Background 26. (Baire category theorem)

If a complete metric space is a countable union of closed sets, then at least one of them contains an open ball.

Proof of Theorem 2.25 (sketch). The sets $E_n = \{x \in X : \sup_\alpha \|T_\alpha x\| \le n\}$ are closed (intersections of closed sets, by continuity of each T_α) and $\bigcup_n E_n = X$ by assumption. By Baire, some E_n contains a ball $B(x_0, r)$. For $\|x\| \le 1$, both x_0 and $x_0 + rx$ lie in E_n, so $\|T_\alpha x\| = \frac{1}{r}\|T_\alpha(x_0 + rx) - T_\alpha x_0\| \le \frac{2n}{r}$ for all α. □

Proof of Theorem 2.23. Let $X := C(\mathbb{T})$ with $\|f\| = \max_{z\in\mathbb{T}} |f(z)|$ (a Banach space), $Y := \mathbb{C}$ and

$$\varphi_N(f) := S_N(f,1) = \sum_{k=-N}^{N} \hat f(k), \qquad f \in C(\mathbb{T}).$$

φ_N is linear, $\varphi_N(\alpha f + \beta g) = \alpha\varphi_N(f) + \beta\varphi_N(g)$, because $\hat f(k) = \left(f, z^k\right)_{L^2}$ is linear in f. It is bounded: $|\hat f(k)| = \left|\int_{\mathbb{T}} f\,\overline{z}^k\,dm\right| \le \max_{\mathbb{T}} |f| \cdot m(\mathbb{T}) = \|f\|$, hence $\|\varphi_N\| \le 2N + 1$.

Suppose $S_N(f,1)$ converged (to a finite limit) for every $f \in C(\mathbb{T})$. Then $\sup_N |\varphi_N(f)| < \infty$ for every f, and Banach–Steinhaus gives $\sup_{N\ge 0} \|\varphi_N\| < \infty$. We show that this is not the case, which proves that there is $f \in C(\mathbb{T})$ with $\sup_N |S_N(f,1)| = \infty$; in particular $S_N(f,1)$ diverges. (Rotating f moves the bad point 1 to any prescribed $z \in \mathbb{T}$.)

Computing $\|\varphi_N\|$. Since $\sum_{k=-N}^{N} \overline{\xi}^k = D_N(-t) = D_N(t)$ for $\xi = e^{it}$ (Remark 2.11),

$$\varphi_N(f) = \sum_{k=-N}^{N} \int_{\mathbb{T}} f(\xi)\overline{\xi}^k\,dm(\xi) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(e^{it}) D_N(t)\,dt.$$

Note D_N is real. In the lecture one takes f_N with $|f_N| = 1$ and $f_N \cdot D_N = |D_N|$, i.e. $f_N = \operatorname{sgn} D_N$; this function is not continuous (it jumps at the zeros of D_N), so we approximate it: $g_j = \frac{D_N}{|D_N| + 1/j}$

is continuous, $|g_j| \le 1$, and $g_j \to \operatorname{sgn} D_N$ wherever $D_N \neq 0$, i.e. except at finitely many points. By dominated convergence,

$$\|\varphi_N\| \ge \lim_{j\to\infty} |\varphi_N(g_j)| = \frac{1}{2\pi}\int_{-\pi}^{\pi} |D_N(t)|\,dt =: L_N \qquad \text{(the } \textit{Lebesgue constants}\text{)}.$$

So it suffices to prove $L_N \to \infty$. Multiplying by $|e^{iNt}| = 1$ and summing the geometric series,

$$\begin{aligned} L_N &= \frac{1}{2\pi}\int_{-\pi}^{\pi} |e^{-iNt} + \cdots + e^{iNt}|dt = \frac{1}{2\pi}\int_{-\pi}^{\pi} |1 + e^{it} + \cdots + e^{2iNt}|dt = \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{|1 - e^{i(2N+1)t}|}{|1 - e^{it}|}dt \\ &\ge \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{|\operatorname{Im}(1 - e^{i(2N+1)t})|}{|1 - e^{it}|}dt = \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{|\sin((2N+1)t)|}{|1 - e^{it}|}dt \ge \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{|\sin((2N+1)t)|}{|t|}dt, \end{aligned}$$

using $|a| \ge |\operatorname{Im} a|$, $\operatorname{Im}(e^{ix}) = \sin x$, and $|1 - e^{it}| \le |t|$ (a chord is shorter than the arc, Fig. 7). By evenness, the substitution $u = (2N+1)t$, and $|\sin u| \ge \sin^2 u = \frac{1-\cos 2u}{2}$,

$$\begin{aligned} L_N &\ge \frac{1}{\pi}\int_0^{(2N+1)\pi} \frac{|\sin u|}{u}\,du \ge \frac{1}{2\pi}\int_1^{(2N+1)\pi} \frac{1 - \cos(2u)}{u}\,du \\ &= \frac{1}{2\pi}\Big(\log((2N+1)\pi) - \int_1^{(2N+1)\pi} \frac{\cos(2u)}{u}\,du\Big). \end{aligned}$$

The last integral stays bounded: integrating by parts, $\int_1^R \frac{\cos 2u}{u}\,du = \left[\frac{\sin 2u}{2u}\right]_1^R + \int_1^R \frac{\sin 2u}{2u^2}du$, which is at most $\frac{1}{2} + \frac{1}{2} + \frac{1}{2}$ in absolute value (it converges to $\int_1^\infty \frac{\cos 2u}{u}\,du$). Hence $L_N \ge \frac{1}{2\pi}(\log((2N+1)\pi) - \frac{3}{2}) \to \infty$. □

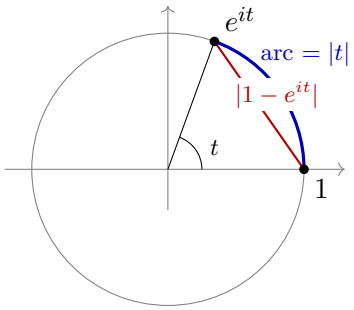


Figure 7: $|1 - e^{it}| = 2|\sin\frac{t}{2}| \le |t|$: the chord is shorter than the arc.

2.6 Functions of bounded variation

Definition 2.27. Let $f\colon \mathbb{T} \to \mathbb{C}$. We say that f has *bounded variation*, $f \in BV(\mathbb{T})$, if

$$V_{\mathbb{T}}(f) := \sup\Big\{\sum_{j=1}^{n} |f(z_{j+1}) - f(z_j)|\Big\} < \infty,$$

the supremum taken over all finite partitions $z_1, \dots, z_n$ of $\mathbb{T}$ (points ordered counter-clockwise, $z_{n+1} := z_1$).

Definition 2.28. Let $g\colon [a,b] \to \mathbb{C}$. Then $g \in BV([a,b])$ if

$$V_a^b(g) := \sup_{a=t_1<t_2<\cdots<t_n=b} \sum_{k=1}^{n-1} |g(t_{k+1}) - g(t_k)| < \infty.$$

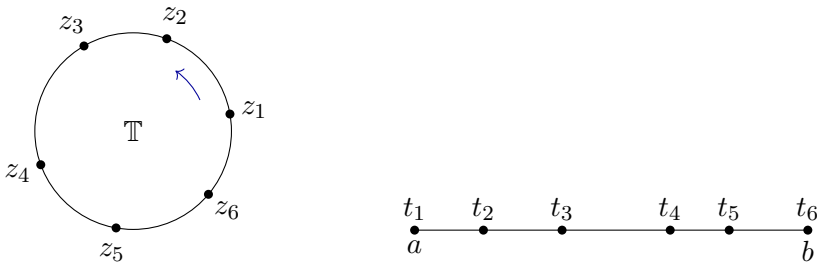


Figure 8: Partitions of $\mathbb{T}$ and of $[a,b]$ used in the definition of the total variation.

Remark 2.29. Let $g(t) = f(e^{it})$. Then $g \in BV([0,2\pi]) \iff f \in BV(\mathbb{T})$, and in fact $V_{\mathbb{T}}(f) = V_0^{2\pi}(g)$. Indeed, adding a point to a partition can only increase the sum (triangle inequality), so in both suprema we may assume the point $1 = e^{i0}$ is included; and partitions of $\mathbb{T}$ containing 1 correspond exactly to partitions $0 = t_1 < \cdots < t_{n+1} = 2\pi$ of $[0,2\pi]$, with the same sums (as $g(2\pi) = g(0)$). Also, BV functions are bounded: $|g(t)| \le |g(a)| + V_a^b(g)$.

Theorem 2.30 (Jordan decomposition)**.** $g \in BV([a,b]) \iff$ *there are bounded nondecreasing functions* $g_1, g_2, g_3, g_4 \colon [a,b] \to \mathbb{R}$ *with*

$$g = g_1 - g_2 + i(g_3 - g_4).$$

Proof. "⇐": a nondecreasing h has $\sum_k |h(t_{k+1}) - h(t_k)| = h(b) - h(a)$ for every partition, and sums of BV functions are BV. "⇒": it suffices to treat real g (apply it to $\operatorname{Re} g$, $\operatorname{Im} g$, whose variation is $\le V(g)$). Let $V(x) = V_a^x(g)$. Then V is nondecreasing, and for $x < y$, $g(y) - g(x) \le V_x^y(g) = V(y) - V(x)$, so $V - g$ is nondecreasing as well; $g = V - (V - g)$. □

Background 31. (Complex measures and Lebesgue–Stieltjes measures)

A *complex measure* on $\mathcal{B}(\mathbb{R})$ is a map $\mu \colon \mathcal{B}(\mathbb{R}) \to \mathbb{C}$ with $\mu(\bigsqcup A_n) = \sum \mu(A_n)$ (absolutely convergent) for disjoint A_n. Its *total variation* is the positive measure $|\mu|(A) = \sup \sum_j |\mu(A_j)|$ (sup over finite partitions of A into Borel sets); it is always finite: $|\mu|(\mathbb{R}) < \infty$, and $\left|\int h\, d\mu\right| \le \int |h|\, d|\mu|$.
Lebesgue–Stieltjes: for every nondecreasing, right-continuous, bounded F on $[a,b]$ there is a unique finite Borel measure μ_F on $(a,b]$ with $\mu_F((a,x]) = F(x) - F(a)$.

Theorem 2.32. $(g \in BV([a,b])$ *and* g *is continuous from the right)* $\iff g(x) = g(a) + \int_{(a,x]} d\mu(s)$ *for all* x*, where* μ *is a complex valued measure on* $(a,b]$ *(of finite total variation).*

Idea. "⇒": write $g = g_1 - g_2 + i(g_3 - g_4)$ as in Theorem 2.30; one can choose the g_j right-continuous when g is, and then $\mu = \mu_{g_1} - \mu_{g_2} + i(\mu_{g_3} - \mu_{g_4})$. "⇐": $\sum_k |g(t_{k+1}) - g(t_k)| = \sum_k |\mu((t_k, t_{k+1}])| \le |\mu|((a,b])$, and right-continuity follows from continuity of μ along decreasing sets $(a, x + \frac{1}{n}] \downarrow (a,x]$. □

Example 2.33 (The Cantor staircase)**.** Let $C = \{\sum_{n\ge1} a_n 3^{-n} : a_n \in \{0,2\}\} \subset [0,1]$ be the Cantor set (remove middle thirds repeatedly); $\lambda(C) = 0$. The *Cantor function* $F(\sum a_n 3^{-n}) = \sum \frac{a_n}{2} 2^{-n}$ on C, extended to be constant on each removed interval, is continuous, nondecreasing, with $F(0) = 0$, $F(1) = 1$. So $F \in BV([0,1])$ and $F(x) = \mu((0,x])$ for the *Cantor measure* μ. Note $F' = 0$ a.e., yet F is not constant: μ has no atoms but lives on the null set C (it is *singular continuous*). Hence μ in Theorem 2.32 cannot in general be replaced by $g'(s)\, ds$.

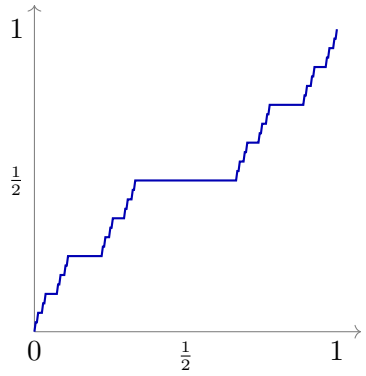


Figure 9: The Cantor staircase (5th stage of the construction): a continuous BV function whose derivative is 0 almost everywhere.

Example 2.34. If $g \in C^1([a,b])$, then $g \in BV([a,b])$:

$$\sum_k |g(t_{k+1}) - g(t_k)| \le c \sum_k |t_{k+1} - t_k| = c\,(b-a) < \infty, \qquad c = \max_{[a,b]} |g'|,$$

by the inequality $|g(x) - g(y)| = \left|\int_y^x g'\right| \le \max_{[a,b]} |g'| \cdot |x-y|$ (for complex g use this integral form, the mean value theorem itself fails for complex valued functions). Here μ in Theorem 2.32 is $d\mu = g'(s)\,ds$: the derivative of such a function "is" a finite measure.

Remark 2.35. If $g \in BV([a,b])$, then for all $x_0 \in [a,b]$ the one-sided limits

$$g(x_0 - 0) = \lim_{x \to x_0,\, x < x_0} g(x), \qquad g(x_0 + 0) = \lim_{x \to x_0,\, x > x_0} g(x)$$

exist. Moreover g has at most countably many discontinuities.

Proof. By Theorem 2.30, $g = g_1 - g_2 + i(g_3 - g_4)$, and for bounded monotone functions $g_k(x_0 \pm 0)$ exist (e.g. $g_k(x_0 - 0) = \sup_{x<x_0} g_k(x)$ for nondecreasing g_k). A nondecreasing h is discontinuous at x iff its jump $h(x+0) - h(x-0) > 0$; since any finitely many jumps sum to at most $h(b) - h(a)$, there are at most $n(h(b) - h(a))$ jumps $> \frac{1}{n}$, so countably many jumps altogether. The discontinuities of g lie in the union of those of $g_1, \dots, g_4$. □

2.7 Hardy's theorem (Dirichlet–Jordan test)

Theorem 2.36 (Hardy)**.** *Let $f \in BV(\mathbb{T})$. Then for every $z \in \mathbb{T}$*

$$\sum_{k=-\infty}^{\infty} \hat{f}(k) z^k := \lim_{N \to \infty} S_N(f, z) = \frac{f(z-0) + f(z+0)}{2}.$$

In particular, if $f \in BV(\mathbb{T})$ is continuous at $z \in \mathbb{T}$, then $\sum_{k=-\infty}^{\infty} \hat{f}(k) z^k = f(z)$.

(This is classically known as the Dirichlet–Jordan theorem; Hardy's contribution is the Tauberian argument below.) The strategy: by Fejér the series is Cesàro summable; if in addition the terms are $O(\frac{1}{k})$, Cesàro convergence upgrades to ordinary convergence. Results of the form "summable in a weak sense + a size condition on the terms ⇒ convergent" are called *Tauberian theorems.*

Lemma 2.37 (Hardy's Tauberian theorem)**.** *Assume that $\sum_{n=1}^{\infty} a_n$ converges in the Cesàro sense and $|a_k| \le \frac{C}{k}$ for all $k \ge 1$ and some $C > 0$. Then $\sum_{k=1}^{\infty} a_k$ converges in the usual sense, to the same sum.*

Proof. Notation:

$$S_n := \sum_{k=1}^{n} a_k, \qquad \sigma_n := \frac{1}{n}\sum_{k=1}^{n} S_k \to \sigma, \qquad \sigma_{n,j} := \frac{S_{n+1}+\cdots+S_{n+j}}{j}, \qquad l_n(\varepsilon) = [\varepsilon n]$$

for $\varepsilon \in (0,1]$, $n \geq 1$, where $[\,\cdot\,]$ is the integer part. We only consider $n \geq 1/\varepsilon$, so that $j = l_n(\varepsilon) \geq 1$; note $\frac{\varepsilon n}{2} \leq j \leq \varepsilon n$. The proof has two steps:

- Step 1: $\lim_{n\to\infty} |\sigma_n - \sigma_{n,l_n(\varepsilon)}| = 0$;
- Step 2: $\limsup_{n\to\infty} |S_n - \sigma_{n,l_n(\varepsilon)}| \leq C\varepsilon$ for every ε.

Step 1 (uses only Cesàro convergence). Since $S_1 + \cdots + S_m = m\sigma_m$,

$$\sigma_{n,j} = \frac{(S_1+\cdots+S_{n+j}) - (S_1+\cdots+S_n)}{j} = \sigma_{n+j}\frac{n+j}{j} - \sigma_n\frac{n}{j} = (\sigma_{n+j}-\sigma_n)\frac{n+j}{j} + \sigma_n.$$

Thus, with $j = l_n(\varepsilon)$,

$$|\sigma_{n,j} - \sigma_n| = |\sigma_{n+j} - \sigma_n| \cdot \frac{n+j}{j} \leq \underbrace{|\sigma_{n+l_n(\varepsilon)} - \sigma_n|}_{\to 0} \cdot \underbrace{\frac{n+\varepsilon n}{\varepsilon n/2}}_{=2\frac{1+\varepsilon}{\varepsilon},\ \text{indep. of } n} \longrightarrow 0,$$

because (σ_m) converges, so $\sigma_{n+j} - \sigma_n \to 0$ for any $j = j(n) \geq 0$.

Step 2 (uses only the size condition). Since $S_n - S_{n+i} = -(a_{n+1} + \cdots + a_{n+i})$,

$$\begin{aligned}\Big|S_n - \frac{S_{n+1}+\cdots+S_{n+j}}{j}\Big| &= \frac{1}{j}\Big|\sum_{i=1}^{j}(S_n - S_{n+i})\Big| \leq \frac{1}{j}\sum_{i=1}^{j}(|a_{n+1}| + \cdots + |a_{n+i}|) \\ &\leq |a_{n+1}| + \cdots + |a_{n+j}| \leq C\Big(\frac{1}{n+1} + \cdots + \frac{1}{n+j}\Big) \\ &\leq \frac{Cj}{n+1} \leq C\,\frac{\varepsilon n}{n+1} \leq C\varepsilon.\end{aligned}$$

Taking the lim sup we obtain Step 2.

Conclusion. $S_n \to \sigma$ iff $\lim |S_n - \sigma| = 0$. By the triangle inequality,

$$\limsup_n |S_n - \sigma| \leq \underbrace{\limsup_n |S_n - \sigma_{n,l_n(\varepsilon)}|}_{\leq C\varepsilon \text{ by Step 2}} + \underbrace{\limsup_n |\sigma_{n,l_n(\varepsilon)} - \sigma_n|}_{=0 \text{ by Step 1}} + \underbrace{\limsup_n |\sigma_n - \sigma|}_{=0 \text{ by Cesàro convergence}},$$

so $\limsup_n |S_n - \sigma| \leq C\varepsilon$ for every $\varepsilon > 0$, i.e. $\lim |S_n - \sigma| = 0$. □

Remark 2.38 (Indexing). Fejér's means are $\frac{1}{N+1}\sum_{n=0}^{N} S_n$ for a series $\sum_{n\geq 0} a_n$, while Lemma 2.37 uses $\frac{1}{n}\sum_{k=1}^{n} S_k$ for $\sum_{n\geq 1} a_n$. This is only a shift of index: putting $b_n = a_{n-1}$ $(n \geq 1)$, the partial sums and means of $\sum b_n$ are S_{n-1} and $\frac{1}{n}\sum_{k=0}^{n-1} S_k$, and $|a_n| \leq \frac{C}{n+1}$ for $n \geq 0$ becomes $|b_n| \leq \frac{C}{n}$.

Lemma 2.39. *If $f \in BV(\mathbb{T})$, then $|\hat{f}(k)| \leq \frac{C}{|k|+1}$ for all $k \in \mathbb{Z}$, with C depending on f only.*

Proof (lecture version, assuming f continuous from the right). Let $g(t) = f(e^{it})$ for $t \in [0, 2\pi]$; then $g \in BV([0,2\pi])$ (remark after the definition of BV). Assume g is continuous from the right (an extra condition). Then, by Theorem 2.32, $g(t) = g(0) + \int_{(0,t]} d\mu(s)$ for a finite complex measure μ. For $k \neq 0$,

$$\begin{aligned}\hat{f}(k) &= \int_{\mathbb{T}} f(\xi)\bar{\xi}^k\, dm(\xi) = \frac{1}{2\pi}\int_0^{2\pi} g(t)e^{-ikt}dt = \frac{1}{2\pi}\int_0^{2\pi}\Big(g(0) + \int_{(0,t]} d\mu(s)\Big)e^{-ikt}dt \\ &= \underbrace{\frac{g(0)}{2\pi}\int_0^{2\pi} e^{-ikt}dt}_{=0 \text{ for } k\neq 0} + \frac{1}{2\pi}\int_0^{2\pi}\int_{(0,2\pi]} \chi_{(0,t]}(s)\, e^{-ikt}\, d\mu(s)\, dt \\ &\overset{\text{Fubini}}{=} \frac{1}{2\pi}\int_{(0,2\pi]}\Big(\int_s^{2\pi} e^{-ikt}dt\Big)d\mu(s),\end{aligned}$$

where Fubini applies since the integrand is bounded and $|\mu|$, dt are finite. Now $\int_s^{2\pi} e^{-ikt}dt = \frac{e^{-ikt}}{-ik}\Big|_s^{2\pi} = \frac{e^{-iks}-1}{ik}$ has modulus $\le \frac{2}{|k|}$, so

$$|\hat f(k)| \le \frac{1}{2\pi}\cdot\frac{2}{|k|}\cdot|\mu|((0,2\pi]) = O\Big(\frac{1}{|k|}\Big).$$

Together with $|\hat f(0)| \le \sup|f|$ this gives the claim, since $\frac{1}{|k|} \le \frac{2}{|k|+1}$ for $k \neq 0$. "In general this assumption is not important, just a longer proof" – here is a short proof without it. □

Proof (general). Extend $g(t) = f(e^{it})$ 2π-periodically; let $V = V_{\mathbb{T}}(f)$. Fix $k \neq 0$ and $h = \frac{\pi}{|k|}$, so $e^{-ik(t+h)} = -e^{-ikt}$. Shifting the variable of integration by h (periodicity), $\hat f(k) = \frac{1}{2\pi}\int_0^{2\pi} g(t+h)e^{-ik(t+h)}dt = -\frac{1}{2\pi}\int_0^{2\pi} g(t+h)e^{-ikt}dt$. Averaging with the original formula,

$$|\hat f(k)| = \Big|\frac{1}{4\pi}\int_0^{2\pi}(g(t)-g(t+h))e^{-ikt}dt\Big| \le \frac{1}{4\pi}\int_0^{2\pi}|g(t+h)-g(t)|\,dt.$$

Since $2\pi = 2|k|\cdot h$, split $[0,2\pi)$ into $2|k|$ intervals of length h:

$$\int_0^{2\pi}|g(t+h)-g(t)|\,dt = \int_0^h\sum_{j=0}^{2|k|-1}|g(s+(j+1)h)-g(s+jh)|\,ds \le h\,V,$$

because for each s the points $e^{i(s+jh)}$ form a partition of $\mathbb{T}$. Hence $|\hat f(k)| \le \frac{hV}{4\pi} = \frac{V}{4|k|}$. □

Proof of Theorem 2.36. Fix $z \in \mathbb{T}$. A BV function is bounded and has one-sided limits $f(z\pm 0)$ at every $z \in \mathbb{T}$ (Remark 2.35). By Fejér's theorem in the form of Remark 2.22, $\sum_{k=-\infty}^{\infty}\hat f(k)z^k$ converges in the Cesàro sense to $\frac{f(z-0)+f(z+0)}{2}$.

Reduction to a continuous-from-the-right f (lecture). Changing f at its points of discontinuity, e.g. replacing it by $\tilde f(e^{it}) = g(t+0)$, does not change $\sum\hat f(k)z^k$ nor $\frac{f(z-0)+f(z+0)}{2}$: the set of discontinuities is countable (Remark 2.35), hence of measure 0, so the integrals $\hat f(k) = \int f\bar\xi^k\,dm$ are unchanged; and one-sided limits are not affected by the values at single points. Also $V_{\mathbb{T}}(\tilde f) \le V_{\mathbb{T}}(f)$ (partition sums of $\tilde f$ are limits of partition sums of f). With the general proof of Lemma 2.39 this reduction is not even needed.

Tauberian step. By Lemma 2.39, $|\hat f(k)| \le \frac{C}{|k|+1}$ for all k. Group the terms symmetrically: $a_0 = \hat f(0)$ and $a_n = \hat f(n)z^n + \hat f(-n)z^{-n}$ for $n \ge 1$. Then $\sum_{n=0}^N a_n = S_N(f,z)$ and $|a_n| \le \frac{2C}{n+1}$. The series $\sum a_n$ converges in the Cesàro sense, so by Lemma 2.37 (with Remark 2.38) it converges in the usual sense to the same value:

$$\sum_{k\in\mathbb{Z}}\hat f(k)z^k = \lim_{N\to\infty}S_N(f,z) = \frac{f(z-0)+f(z+0)}{2}.$$ □

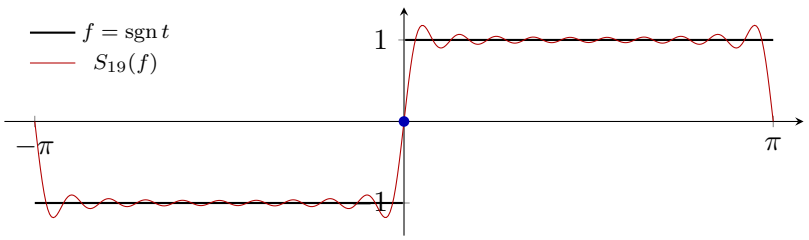


Figure 10: Hardy's theorem for the BV function $f(e^{it}) = \operatorname{sgn} t$: at the jump $t = 0$ every partial sum equals $0 = \frac{f(1-0)+f(1+0)}{2}$; at points of continuity $S_N(f) \to f$.

3 Exercises

The problem sheets of the course, with detailed solutions. The sheets write $\langle \cdot, \cdot \rangle$ for the inner product and χ_A for the indicator function of a set A; we keep their notation here, so $\langle h, e \rangle = (h, e)$ in the notation of the lectures.

3.1 List 1 (13. 10. 2026)

Problem 1. Let H be a (complex, separable) Hilbert space.

(a) Let $(e_i)_{i \in I}$ be an ONS (orthonormal system). Using separability, prove that I is at most countable. *Hint.* Show that $\|e_i - e_j\| = \sqrt{2}$ for $i \neq j$, and use a countable dense subset of H.

(b) Let $e_1, \dots, e_N$ be orthonormal, and put

$$V_N = \operatorname{span}\{e_1, \dots, e_N\}, \qquad P_N h = \sum_{k=1}^{N} \langle h, e_k \rangle e_k.$$

Verify that $P_N h$ is the orthogonal projection of h onto V_N, and then use the Pythagorean theorem to deduce Bessel's inequality: $\sum_{k=1}^{\infty} |\langle h, e_k \rangle|^2 \leq \|h\|^2$.

(c) Let $(e_n)_{n=1}^{\infty}$ be an ONS. Prove that the following are equivalent:

(i) $(e_n)_{n=1}^{\infty}$ is an ONB;
(ii) $\langle h, e_k \rangle = 0$ for all $k \geq 1$ implies $h = 0$;
(iii) for every $h \in H$, $h = \sum_{k=1}^{\infty} \langle h, e_k \rangle e_k$ in H;
(iv) for every $h \in H$, $\|h\|^2 = \sum_{k=1}^{\infty} |\langle h, e_k \rangle|^2$.

Hint. For (iv)⇒(i), you may use the Pythagorean theorem.

Solution. **(a)** This is Exercise 1.22; we repeat the argument for an arbitrary ONS (completeness is not needed). For $i \neq j$, by orthogonality (Pythagoras),

$$\|e_i - e_j\|^2 = \langle e_i - e_j, e_i - e_j \rangle = \|e_i\|^2 - \langle e_i, e_j \rangle - \langle e_j, e_i \rangle + \|e_j\|^2 = 1 - 0 - 0 + 1 = 2.$$

Let $D \subset H$ be countable and dense (it exists since H is separable, Theorem 1.21). For each $i \in I$ the open ball $B_i = B(e_i, \frac{\sqrt{2}}{2})$ contains some point of D (density); choose one and call it d_i. If $d_i = d_j$ for $i \neq j$, then $\|e_i - e_j\| \leq \|e_i - d_i\| + \|d_j - e_j\| < \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} = \sqrt{2}$, a contradiction. So $i \mapsto d_i$ is an injective map $I \to D$, and I is at most countable.

(b) *Orthogonal projection.* Recall that the orthogonal projection of h onto a subspace V is the (unique) vector $v \in V$ with $h - v \perp V$, i.e. $\langle h - v, w \rangle = 0$ for all $w \in V$. Uniqueness: if v, v' both have this property then $v - v' \in V$ and $v - v' = (h - v') - (h - v) \perp V$, so $\langle v - v', v - v' \rangle = 0$.

Clearly $P_N h \in V_N$. For $j \leq N$, by orthonormality,

$$\langle h - P_N h, e_j \rangle = \langle h, e_j \rangle - \sum_{k=1}^{N} \langle h, e_k \rangle \langle e_k, e_j \rangle = \langle h, e_j \rangle - \langle h, e_j \rangle = 0,$$

and by linearity (and conjugate linearity in the second slot) $\langle h - P_N h, w \rangle = 0$ for every $w \in V_N$. Hence $P_N h$ is the orthogonal projection of h onto V_N. It is also the point of V_N closest to h: for $v \in V_N$, $P_N h - v \in V_N$, so by Pythagoras $\|h - v\|^2 = \|h - P_N h\|^2 + \|P_N h - v\|^2 \geq \|h - P_N h\|^2$.

Bessel. Since $h = P_N h + (h - P_N h)$ with $P_N h \perp h - P_N h$, the Pythagorean theorem gives

$$\|h\|^2 = \|P_N h\|^2 + \|h - P_N h\|^2 \geq \|P_N h\|^2 = \Big\|\sum_{k=1}^{N} \langle h, e_k\rangle e_k\Big\|^2 = \sum_{k=1}^{N} |\langle h, e_k\rangle|^2,$$

the last equality again by orthonormality. Now let $e_1, e_2, \ldots$ be an infinite ONS: the inequality holds for every N, the partial sums of $\sum_k |\langle h, e_k\rangle|^2$ are nondecreasing and bounded by $\|h\|^2$, so the series converges and $\sum_{k=1}^{\infty} |\langle h, e_k\rangle|^2 \leq \|h\|^2$. (Compare with the proof of Theorem 1.26, Step 1, where the same inequality was obtained from Cauchy–Schwarz.)

(c) Throughout, $P_N h = \sum_{k=1}^{N} \langle h, e_k\rangle e_k$ as in (b), and we use $\|h\|^2 = \|P_N h\|^2 + \|h - P_N h\|^2$ and $\|P_N h\|^2 = \sum_{k \leq N} |\langle h, e_k\rangle|^2$.

(i)⇒(ii). Let $\langle h, e_k\rangle = 0$ for all k. Then $\langle h, w\rangle = 0$ for all $w \in \operatorname{span}\{e_k\}$. If $w_n \to w$, then $\langle h, w\rangle = \lim \langle h, w_n\rangle = 0$ (continuity of the inner product, Lemma 1.28); so $\langle h, w\rangle = 0$ for all $w \in \operatorname{clos}\operatorname{span}\{e_k\} = H$. Take $w = h$: $\|h\|^2 = 0$, so $h = 0$.

(ii)⇒(iii). Let $h \in H$. For $N > M$, $\|P_N h - P_M h\|^2 = \sum_{k=M+1}^{N} |\langle h, e_k\rangle|^2$, which tends to 0 as $M \to \infty$ by Bessel's inequality (tails of a convergent series). So $(P_N h)$ is Cauchy, and since H is complete it has a limit $\tilde{h} = \sum_{k=1}^{\infty} \langle h, e_k\rangle e_k$. For fixed k and $N \geq k$, $\langle P_N h, e_k\rangle = \langle h, e_k\rangle$; letting $N \to \infty$, $\langle \tilde{h}, e_k\rangle = \langle h, e_k\rangle$. Thus $\langle h - \tilde{h}, e_k\rangle = 0$ for all k, and (ii) gives $h = \tilde{h}$.

(iii)⇒(iv). By (iii), $P_N h \to h$, hence $\|P_N h\| \to \|h\|$ (Lemma 1.28), i.e. $\sum_{k=1}^{N} |\langle h, e_k\rangle|^2 = \|P_N h\|^2 \to \|h\|^2$.

(iv)⇒(i). The system is orthonormal by assumption; we must show completeness. By Pythagoras and (iv),

$$\|h - P_N h\|^2 = \|h\|^2 - \|P_N h\|^2 = \|h\|^2 - \sum_{k=1}^{N} |\langle h, e_k\rangle|^2 \xrightarrow{N\to\infty} 0.$$

So $h = \lim P_N h$ is a limit of elements of $\operatorname{span}\{e_k\}$, i.e. $h \in \operatorname{clos}\operatorname{span}\{e_k\}$. As h was arbitrary, $\operatorname{clos}\operatorname{span}\{e_k\} = H$ and (e_n) is an ONB.

Remark 3.1. Problem 1(c) is a reformulation of the Riesz–Fischer theorem (Theorem 1.26) together with Corollary 1.29: (i)⇒(iii),(iv) is Theorem 1.26, and (ii) is the practical test for completeness.

Problem 2. Let (X, μ) be a finite measure space.

(a) Prove that $L^2(X,\mu) \subseteq L^1(X,\mu)$ and that $\|f\|_{L^1} \leq \mu(X)^{1/2} \|f\|_{L^2}$. In particular, explain why the Fourier coefficients of functions in $L^2(\mathbb{T}, m)$ are well defined, where m is the normalized arc length measure defined in the lectures, given by $\int_{\mathbb{T}} f(z)\, dm(z) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(e^{ix})\, dx$.

(b) For $\alpha \geq 0$ consider $f_\alpha(x) = |x|^{-\alpha}$, $x \in (-1,1) \setminus \{0\}$. Determine the values of α for which $f_\alpha \in L^1(-1,1)$.

(c) Let $f \in L^2(X,\mu)$ and set $f_M = f \cdot \chi_{\{|f| \leq M\}}$, $M > 0$. Prove that $f_M \to f$ in $L^2(X,\mu)$ as $M \to \infty$. *Hint.* Apply DCT (the dominated convergence theorem) to $|f - f_M|^2$.

(d) On $(0,1)$ with Lebesgue measure, consider $f_n = \sqrt{n}\, \chi_{(0,1/n)}$. Show that $f_n \to 0$ almost everywhere but not in L^2. Explain why DCT cannot be used here to conclude L^2 convergence.

Background 2. (L^1 and the convergence theorems)

$L^1(X,\mu)$ is the space of (classes of a.e. equal) measurable f with $\|f\|_{L^1} = \int_X |f|\, d\mu < \infty$; for such f the integral $\int_X f\, d\mu \in \mathbb{C}$ is defined and $\left|\int f\, d\mu\right| \leq \int |f|\, d\mu$.
Monotone convergence theorem (MCT). If $0 \leq g_1 \leq g_2 \leq \ldots$ are measurable and $g_n \to g$ pointwise, then $\int g_n\, d\mu \to \int g\, d\mu$ (possibly $= \infty$).
Dominated convergence theorem (DCT). If $g_n \to g$ μ-a.e. and there is an integrable $G \geq 0$ (independent of n) with $|g_n| \leq G$ a.e. for all n, then $g \in L^1$ and $\int |g_n - g|\, d\mu \to 0$; in particular

$\int g_n \to \int g$.

Solution. **(a)** Let $f \in L^2(X,\mu)$. The constant function 1 is in $L^2(X,\mu)$ since $\int_X 1^2\,d\mu = \mu(X) < \infty$. By Cauchy–Schwarz in L^2 (Theorem 1.7) applied to $|f|$ and 1,

$$\|f\|_{L^1} = \int_X |f| \cdot 1\,d\mu = \langle |f|, 1\rangle_{L^2} \le \|f\|_{L^2}\,\|1\|_{L^2} = \mu(X)^{1/2}\,\|f\|_{L^2} < \infty.$$

So $f \in L^1$. (Functions equal a.e. have the same L^1 and L^2 norms, so this is a statement about classes, and the inclusion map $L^2 \to L^1$ is a bounded linear map.)

Fourier coefficients. For $f \in L^2(\mathbb{T}, m)$ and $k \in \mathbb{Z}$, the function $z \mapsto f(z)\overline{z}^k$ is measurable and $|f(z)\overline{z}^k| = |f(z)|$ since $|z| = 1$. As $m(\mathbb{T}) = 1$, $\int_{\mathbb{T}} |f\overline{z}^k|\,dm = \|f\|_{L^1} \le \|f\|_{L^2} < \infty$, so

$$\hat{f}(k) = \int_{\mathbb{T}} f(z)\overline{z}^k\,dm(z) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(e^{ix})e^{-ikx}dx$$

is the integral of an integrable function, hence a well defined complex number, with $|\hat{f}(k)| \le \|f\|_{L^1} \le \|f\|_{L^2}$. It does not depend on the representative of f (changing f on a null set does not change the integral). Note also $\hat{f}(k) = \langle f, z^k\rangle_{L^2}$, which is finite by Cauchy–Schwarz directly. (Since $m(\mathbb{T}) = 1$ the coefficients make sense even for $f \in L^1(\mathbb{T}, m)$, a larger space by (b).)

(b) $f_\alpha \ge 0$ is even, so $\int_{-1}^1 f_\alpha = 2\int_0^1 x^{-\alpha}dx$. By MCT, $\int_0^1 x^{-\alpha}dx = \lim_{\varepsilon\downarrow 0}\int_\varepsilon^1 x^{-\alpha}dx$, and

$$\int_\varepsilon^1 x^{-\alpha}dx = \begin{cases} \dfrac{1-\varepsilon^{1-\alpha}}{1-\alpha} \xrightarrow{\varepsilon\to 0} \dfrac{1}{1-\alpha}, & \alpha < 1, \\ -\log\varepsilon \xrightarrow{\varepsilon\to 0} \infty, & \alpha = 1, \\ \dfrac{\varepsilon^{1-\alpha}-1}{\alpha-1} \xrightarrow{\varepsilon\to 0} \infty, & \alpha > 1. \end{cases}$$

Hence $f_\alpha \in L^1(-1,1)$ if and only if $0 \le \alpha < 1$, and then $\|f_\alpha\|_{L^1} = \frac{2}{1-\alpha}$.

Comment. Since $f_\alpha^2 = f_{2\alpha}$, we get $f_\alpha \in L^2(-1,1) \iff \alpha < \frac{1}{2}$. So for $\frac{1}{2} \le \alpha < 1$, $f_\alpha \in L^1 \setminus L^2$: the inclusion in (a) is strict. On spaces of infinite measure there is no inclusion at all, e.g. $\frac{1}{1+|x|} \in L^2(\mathbb{R}) \setminus L^1(\mathbb{R})$.

(c) f_M is measurable ($\{|f| \le M\}$ is a measurable set) and

$$|f - f_M|^2 = |f|^2\,\chi_{\{|f|>M\}}.$$

Pointwise convergence. Since $\int |f|^2 d\mu < \infty$, $|f(x)| < \infty$ for a.e. x. For such x, as soon as $M \ge |f(x)|$ we have $\chi_{\{|f|>M\}}(x) = 0$; hence $|f - f_M|^2(x) \to 0$ as $M \to \infty$, for a.e. x.

Domination. $|f - f_M|^2 \le |f|^2$, and $|f|^2 \in L^1$ because $f \in L^2$.

By DCT (along any sequence $M_n \to \infty$), $\|f - f_M\|_{L^2}^2 = \int_X |f - f_M|^2\,d\mu \to 0$. Since $M \mapsto \|f - f_M\|_{L^2}$ is nonincreasing, convergence along sequences implies convergence as $M \to \infty$ through all reals. (Finiteness of μ is not needed here.)

Meaning. Bounded functions are dense in L^2: every f is approximated by its truncations f_M, $|f_M| \le M$.

(d) *Pointwise convergence.* Fix $x \in (0,1)$. For $n > \frac{1}{x}$ we have $x \notin (0, \frac{1}{n})$, so $f_n(x) = 0$. Hence $f_n(x) \to 0$ for *every* $x \in (0,1)$, in particular almost everywhere.

No L^2 convergence. $\|f_n\|_{L^2}^2 = \int_0^{1/n}(\sqrt{n})^2 dx = n \cdot \frac{1}{n} = 1$, so $\|f_n - 0\|_{L^2} = 1 \not\to 0$. Moreover (f_n) has no L^2-limit at all: if $f_n \to g$ in L^2, a subsequence would converge to g a.e. (a standard fact), so $g = 0$ a.e., contradicting $\|f_n\| = 1$.

Why DCT fails. To apply DCT to $|f_n - 0|^2 = n\chi_{(0,1/n)}$ we would need an integrable G with $n\chi_{(0,1/n)} \le G$ for all n. The smallest candidate is $G = \sup_n n\chi_{(0,1/n)}$. For $x \in [\frac{1}{n+1}, \frac{1}{n})$ we have $x \in (0, \frac{1}{k})$ exactly for $k \le n$, so $G(x) = n > \frac{1}{x} - 1$. Thus

$$\int_0^1 G\,dx \ge \int_0^1 \Big(\frac{1}{x} - 1\Big)dx = \infty,$$

and no integrable dominating function exists. The mass $\int f_n^2 = 1$ does not disappear; it concentrates in the shrinking interval $(0, \frac{1}{n})$ (it "escapes to a point"), which pointwise convergence cannot see.

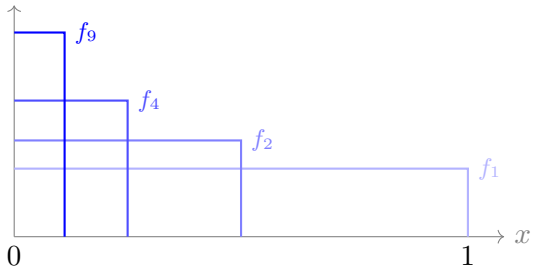


Figure 11: $f_n = \sqrt{n}\,\chi_{(0,1/n)}$: higher and narrower bumps, each with $\int f_n^2 = 1$.

Problem 3. Let $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$.

(a) Show that $\{z^k : k \in \mathbb{Z}\}$ is an ONS in $L^2(\mathbb{T}, m)$, where m is the normalized arc length measure defined above.

(b) Let $I \subset \mathbb{T}$ be a closed arc of positive length strictly less than 2π, i.e. $I = \{e^{it} : x_0 \le t \le x_1\}$, $0 < x_1 - x_0 < 2\pi$. Construct continuous functions φ_ε on $\mathbb{T}$ such that $0 \le \varphi_\varepsilon \le 1$ and $\varphi_\varepsilon \to \chi_I$ in $L^2(\mathbb{T}, m)$ $(\varepsilon \to 0^+)$. *Hint.* We only need to modify χ_I near the endpoints of I.

(c) (optional) Prove that for every measurable set $A \subseteq \mathbb{T}$ and every $\varepsilon > 0$ there is a finite union of arcs $U \subseteq \mathbb{T}$ such that $m(A \triangle U) < \varepsilon$. In particular, $\|\chi_U - \chi_A\|_{L^2}^2 < \varepsilon$.

(d) Deduce that $C(\mathbb{T})$ is dense in $L^2(\mathbb{T}, m)$.

(e) Let $\mathcal{P} = \left\{\sum_{k=-N}^{N} a_k z^k : N \in \mathbb{N},\ a_k \in \mathbb{C}\right\}$. Apply the complex Stone–Weierstrass theorem to show that $\mathcal{P}$ is uniformly dense in $C(\mathbb{T})$. Conclude that $\{z^k : k \in \mathbb{Z}\}$ is an ONB of $L^2(\mathbb{T}, m)$.

This problem is a detailed version of Exercise 2.1, where the two density facts were only quoted.

Solution. **(a)** For $k, j \in \mathbb{Z}$, using the parametrization $z = e^{ix}$,

$$\langle z^k, z^j \rangle = \int_{\mathbb{T}} z^k \bar{z}^j \, dm = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{i(k-j)x} dx = \begin{cases} 1, & k = j, \\ \dfrac{1}{2\pi}\left[\dfrac{e^{i(k-j)x}}{i(k-j)}\right]_{-\pi}^{\pi} = 0, & k \neq j, \end{cases}$$

since $e^{i(k-j)x}$ is 2π-periodic. Each z^k is bounded, hence in L^2. So $\{z^k\}$ is an ONS.

(b) Write $\ell = x_1 - x_0 \in (0, 2\pi)$ and let $0 < \varepsilon < \frac{2\pi - \ell}{2}$, so that the two "ramps" added below do not overlap on the other side of the circle. Define a function ψ_ε on $[x_0 - \varepsilon,\ x_0 - \varepsilon + 2\pi)$ (one full turn) by

$$\psi_\varepsilon(t) = \begin{cases} \dfrac{t - (x_0 - \varepsilon)}{\varepsilon}, & x_0 - \varepsilon \le t \le x_0, \\ 1, & x_0 \le t \le x_1, \\ \dfrac{(x_1 + \varepsilon) - t}{\varepsilon}, & x_1 \le t \le x_1 + \varepsilon, \\ 0, & x_1 + \varepsilon \le t < x_0 - \varepsilon + 2\pi, \end{cases}$$

and set $\varphi_\varepsilon(e^{it}) = \psi_\varepsilon(t)$. The pieces agree at the break points, and ψ_ε tends to 0 at both ends of the period interval, so φ_ε is a well-defined continuous function on $\mathbb{T}$ with $0 \le \varphi_\varepsilon \le 1$ and $\varphi_\varepsilon = \chi_I$

outside the two arcs $J_\varepsilon^- = \{e^{it} : x_0 - \varepsilon < t < x_0\}$ and $J_\varepsilon^+ = \{e^{it} : x_1 < t < x_1 + \varepsilon\}$. On these arcs $|\varphi_\varepsilon - \chi_I| \le 1$. Hence

$$\|\varphi_\varepsilon - \chi_I\|_{L^2}^2 = \int_{J_\varepsilon^- \cup J_\varepsilon^+} |\varphi_\varepsilon - \chi_I|^2 \, dm \le m(J_\varepsilon^-) + m(J_\varepsilon^+) = \frac{2\varepsilon}{2\pi} = \frac{\varepsilon}{\pi} \xrightarrow{\varepsilon \to 0^+} 0.$$

(The same construction works for open or half-open arcs, since endpoints have measure 0; for the full circle $\chi_{\mathbb{T}} = 1$ is already continuous.)

(c) We use the definition of Lebesgue measure through the outer measure: in the parametrization $t \in [-\pi, \pi)$,

$$m(A) = \inf\Big\{\sum_{j=1}^{\infty} m(J_j) : \ J_j \text{ open arcs}, \ A \subseteq \bigcup_j J_j\Big\} \qquad \text{for measurable } A \subseteq \mathbb{T}.$$

Let $\varepsilon > 0$. Choose open arcs $J_1, J_2, \dots$ covering A with $\sum_j m(J_j) < m(A) + \frac{\varepsilon}{2}$, and put $G = \bigcup_j J_j \supseteq A$. Then

$$m(G \setminus A) = m(G) - m(A) \le \sum_j m(J_j) - m(A) < \tfrac{\varepsilon}{2}.$$

As $\sum_j m(J_j) < \infty$, choose n with $\sum_{j>n} m(J_j) < \frac{\varepsilon}{2}$ and let $U = J_1 \cup \cdots \cup J_n$, a finite union of arcs. Since $A \triangle U = (A \setminus U) \cup (U \setminus A)$ is a disjoint union,

$$m(A \triangle U) = m(A \setminus U) + m(U \setminus A) \le m\Big(\bigcup_{j>n} J_j\Big) + m(G \setminus A) < \tfrac{\varepsilon}{2} + \tfrac{\varepsilon}{2} = \varepsilon,$$

because $A \setminus U \subseteq G \setminus U \subseteq \bigcup_{j>n} J_j$ and $U \setminus A \subseteq G \setminus A$. Finally, $|\chi_U - \chi_A| = \chi_{A \triangle U}$ (it equals 1 exactly at points lying in one of the sets but not the other), so

$$\|\chi_U - \chi_A\|_{L^2}^2 = \int_{\mathbb{T}} \chi_{A \triangle U} \, dm = m(A \triangle U) < \varepsilon.$$

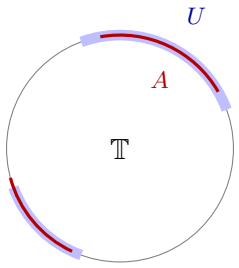


Figure 12: A measurable set A (here two arcs, but in general arbitrary) and a finite union of arcs U with small symmetric difference $A \triangle U$.

(d) Let $\overline{C}$ denote the closure of $C(\mathbb{T})$ in $L^2(\mathbb{T}, m)$; it is a closed linear subspace (closure of a subspace). We show $\overline{C} = L^2$.

Step 1: indicators of arcs. If I is an arc of length $< 2\pi$, then $\chi_I \in \overline{C}$ by (b) (open, closed or half-open arcs differ by at most two points, a null set). For $I = \mathbb{T}$, $\chi_{\mathbb{T}} = 1 \in C(\mathbb{T})$.

Step 2: indicators of finite unions of arcs. A finite union of arcs U can be written as a disjoint union of finitely many arcs $I_1, \dots, I_r$ (up to finitely many points): cut $\mathbb{T}$ at all endpoints and keep the pieces lying in U. Then $\chi_U = \sum_i \chi_{I_i}$ a.e., which lies in $\overline{C}$ by Step 1, as $\overline{C}$ is a linear subspace.

Step 3: indicators of measurable sets. By (c), χ_A is an L^2-limit of functions χ_U from Step 2, so $\chi_A \in \overline{C}$ since $\overline{C}$ is closed.

Step 4: simple functions. A simple function $s = \sum_{j=1}^{r} c_j \chi_{A_j}$ ($c_j \in \mathbb{C}$, A_j measurable) lies in $\overline{C}$ by linearity.

Step 5: all of L^2. Let $f \in L^2$. Writing $f = (u^+ - u^-) + i(v^+ - v^-)$ with $u = \operatorname{Re} f$, $v = \operatorname{Im} f$, each of the four non-negative parts is the pointwise limit of a nondecreasing sequence of simple functions (e.g. $s_n = \min(n, 2^{-n}\lfloor 2^n u^+ \rfloor)$). This gives simple functions $s_n \to f$ pointwise a.e. with $|s_n| \le |f|$ (the standard construction can be arranged so that each part is bounded by the corresponding part of f). Then $|f - s_n|^2 \le (2|f|)^2 = 4|f|^2 \in L^1$ and $|f - s_n|^2 \to 0$ a.e., so by DCT $\|f - s_n\|_{L^2} \to 0$. Hence $f \in \overline{C}$.

Therefore $C(\mathbb{T})$ is dense in $L^2(\mathbb{T}, m)$.

(e) *Stone–Weierstrass* (Theorem 2.2): $\mathbb{T}$ is a compact metric space, and $\mathcal{P} \subset C(\mathbb{T})$ satisfies:

- $\mathcal{P}$ is a subalgebra: it is clearly a linear subspace, and the product of two trigonometric polynomials is one, since $z^k z^j = z^{k+j}$;
- $\mathcal{P}$ contains the constants ($N = 0$);
- $\mathcal{P}$ separates points: $p(z) = z$ belongs to $\mathcal{P}$ and is injective;
- $\mathcal{P}$ is closed under complex conjugation: on $\mathbb{T}$ we have $\overline{z} = z^{-1}$, so $\overline{\sum_{k=-N}^{N} a_k z^k} = \sum_{k=-N}^{N} \overline{a_k} z^{-k} = \sum_{k=-N}^{N} \overline{a_{-k}} z^k \in \mathcal{P}$.

(The last condition is essential: the algebra of polynomials $\sum_{k \ge 0} a_k z^k$ satisfies the first three but is not dense in $C(\mathbb{T})$ – e.g. $\overline{z}$ is not a uniform limit of them, since $\int_{\mathbb{T}} p(z)\, z\, dm = 0$ for such p, while $\int_{\mathbb{T}} \overline{z}\, z\, dm = 1$.) Hence $\mathcal{P}$ is dense in $(C(\mathbb{T}), \|\cdot\|_\infty)$.

ONB. By (a), $\{z^k\}$ is an ONS. Let $f \in L^2(\mathbb{T}, m)$ and $\varepsilon > 0$. By (d) there is $g \in C(\mathbb{T})$ with $\|f - g\|_{L^2} < \varepsilon$, and by the above there is $p \in \mathcal{P}$ with $\|g - p\|_\infty < \varepsilon$. As $m(\mathbb{T}) = 1$, $\|g - p\|_{L^2} = \left(\int |g - p|^2 dm\right)^{1/2} \le \|g - p\|_\infty$, hence

$$\|f - p\|_{L^2} \le \|f - g\|_{L^2} + \|g - p\|_{L^2} < 2\varepsilon.$$

Since $p \in \operatorname{span}\{z^k : k \in \mathbb{Z}\}$, this shows $\operatorname{clos}\operatorname{span}\{z^k\} = L^2(\mathbb{T}, m)$, i.e. $\{z^k : k \in \mathbb{Z}\}$ is an ONB of $L^2(\mathbb{T}, m)$. (Alternatively, use Fejér's theorem instead of Stone–Weierstrass, Corollaries 2.18 and 2.19.)